Foundation Model-Based Spectral–Spatial Transformer for Hyperspectral Image Classification

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***Abstract*— Recently, deep learning models have dominated hyperspectral image (HSI) classification. Nowadays, deep learning is undergoing a paradigm shift with the rise of transformer-based foundation models. In this study, the potential of transformer-based foundation models, including the vision foundation model (VFM) and language foundation model (LFM), for HSI classification are investigated. First, to improve the performance of traditional HSI classification tasks, a spectral–spatial VFM-based transformer (SS-VFMT) is proposed, which inserts spectral–spatial information into the pretrained foundation transformer. Specifically, a given pretrained transformer receives HSI patch tokens for long-range feature extraction benefiting from the prelearned weights. Meanwhile, two enhancement modules, i.e., spatial and spectral enhancement modules (SpaEMs\SpeEMs), utilize spectral and spatial information for steering the behavior of the transformer. Besides, an additional patch relationship distillation strategy is designed for SS-VFMT to exploit the pretrained knowledge better, leading to the proposed SS-VFMT-D. Second, based on SS-VFMT, to address a new HSI classification task, i.e., generalized zero-shot classification, a spectral–spatial vision-language-based transformer (SS-VLFMT) is proposed. This task is to recognize novel classes not seen during training, which is more meaningful as the real world is usually open. The SS-VLFMT leverages SS-VFMT to extract spectral–spatial features and corresponding hash codes while integrating a pretrained language model to extract text features from class names. Experimental results on HSI datasets reveal that the proposed methods are competitive compared to the state-of-the-art methods. Moreover, the foundation model-based methods open a new window for HSI classification tasks, especially for HSI zero-shot classification.**

***Index Terms*— Deep learning, foundation model, hyperspectral image (HSI) classification, transformer, zero-shot learning (ZSL).**

I. Introduction

HYPERSPECTRAL image (HSI) comprises hundreds of nearly contiguous spectral bands covering a wide range of electromagnetic spectrum [1]. In addition to capturing spatial information of land-covers, HSI provides rich spectral information and thus has emerged as a powerful technique for analyzing and understanding the Earth's surface. It has been widely used in various applications such as environmental monitoring [2], agriculture [3], and mineral exploration [4].

Among all the HSI processing technologies, HSI classification is capable of pixel-level category labeling and analysis [5]. Due to the fact that it is a basic technique of HSI processing and applications, HSI classification has attracted lots of attention in the remote sensing community for many years [6], [7].

In early research on HSI classification, a common approach involved integrating classifiers with feature extraction techniques. Popular classifiers included support vector machines and random forests, while feature extraction techniques such as the principal component analysis (PCA) [8], extended morphological profiles (EMPs) [9], and sparse representations [10] were commonly used. However, these methods relied on human-crafted features, requiring careful design and catering to specific scenarios. Besides, their ability to extract high-level image features is restricted. Therefore, these methods usually fail to obtain better classification performance [11].

The remarkable feature extraction capabilities of deep learning methods have made them increasingly popular in the field of HSI classification [12]. Deep learning aims to build neural networks that simulate the process of analysis and learning in the human brain. These networks typically consist of multiple layers, where low-level features are extracted and combined to form more abstract and complex high-level features [13]. Common deep learning networks include deep belief networks (DBNs) [14], stacked autoencoders (SAEs) [15], and recurrent neural networks (RNNs) [16]. Among them, convolutional neural networks (CNNs) [17] and transformers [18] are two kinds of mainstream deep learning models that have gained lots of attention. CNNs use local connection and weight sharing mechanisms, which not only reduce CNNs' number of weights but also endow the model's powerful capability to extract the local features of the inputs [19]. In real applications, HSI images are with the characteristic of locality, and the number of HSI training samples is usually limited [20]. As a result, CNN-based models have been widely explored for HSI classification. Various CNN architectures, including 1-D CNN [21], 2-D CNN [22], 1-D–2-D CNN [23], 3-D CNN [24], and 2-D–3-D CNN [25], have been proposed for spectral and spatial feature extraction. Compared with CNNs which mainly focus on modeling locality, transformers perform well at capturing long-range dependencies in data, allowing for a global understanding of the spatial and spectral relationships in HSIs [26]. Thus, transformer-based HSI classification methods have emerged and obtained superior performance. Meanwhile,

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lots of works focus on integrating the convolution operations into the transformer to take advantage of both model architectures effectively [27], [28].

At present, in the field of artificial intelligence, the improvements in computer hardware, the development of the transformer model architecture, and the availability of much more training data make foundation models possible, powerful, and prevalent [29]. Transformer-based foundation models, such as ChatGPT [30] and DINO [31], have shown their power in many applications and have provided a glimpse of artificial general intelligence. Foundation models, which are usually trained on a dataset at scale, are capable of a wide range of downstream tasks. In many research areas, fine-tuned foundation models achieve significantly better performance compared to task-specific models. For example, DINOv2, which was proposed by Meta AI Research, was pretrained on 1.2B unique images and obtained good results in many computer vision tasks including image classification and semantic segmentation.

In the field of CV, transformer-based foundation models refer to pretrained vision transformers that capture general visual representations that can be fine-tuned or adapted for specific computer vision tasks like object detection, image classification, and semantic segmentation [32], [33]. These models are typically trained on large-scale image datasets using self-supervised learning, and thus, they can capture rich visual features and semantics, which can also potentially improve the HSI classification accuracy. Moreover, the pretrained transformer is endowed with inductive bias through large-scale pretraining, such as locality, which is beneficial for the HSI classification tasks. As for the large language models like Llama [34], they have learned tremendous knowledge and can offer strong language generation and zero-shot transfer abilities. Besides, vision-language foundation model (LFM) research has also witnessed rapid advancement in the past few years [35], [36], which can be adapted to a wider range of downstream tasks [37].

We hope that the focusing foundation models can bring more benefits to HSI classification, such as extending traditional classification tasks to more challenging but more realistic scenarios. Most existing methods for HSI classification are operated under the assumption that all classes present in the test set are also included in the training set [38]. However, the real world is usually open because of the diversity of land covers [39]. When a test sample belongs to an unknown class not seen during training, the existing HSI classification methods tend to misclassify it as a known class from the training data [40]. Therefore, we also want to accomplish generalized zero-shot learning (ZSL) in the HSI classification task, which is to recognize novel classes that are not seen during training. Generally speaking, not only do we want to improve the classification performance of traditional HSI classification using pretrained foundation visual models, but we also want to use LFMs as a bridge to an initial exploration of generalized zero-shot classification of HSI.

However, there are significant challenges when adapting the transformer-based foundation model to the HSI classification task.

1) The foundation models are obtained using natural RGB images. However, there are great differences between RGB images and HSI images. For example, compared to traditional RGB images, HSIs have much higher dimensionality, often spanning hundreds of spectral bands. Besides, the land-covers in an HSI are usually smaller in scale and densely distributed. HSIs also contain many categories of objects that are not recorded in natural images. In summary, special modifications to the foundation model are required to effectively capture relevant spectral and spatial information of HSI classification.
2) The foundation models are usually large in scale for capturing general features using large amounts of training data. Therefore, the overfitting problem occurs when using the limited training HSI samples to adapt the foundation transformer to the HSI classification task. What's more, fully tuning a foundation transformer usually needs more computational resources, which also hinders the application of the foundation model.
3) Directly aligning the visual features with text features from class names makes the decision boundaries overfitting to the seen classes during training. The visual features are generated by the vision encoder using samples of seen classes, and they may lose discriminative information about unseen classes. They also cannot be consistent with the text features obtained by the language model, which is given by prelearned knowledge.

To tackle the above two problems, this article proposes a spatial–spectral foundation model-based transformer for HSI classification, leveraging the pretrained representations to improve classification accuracy. First, PCA is utilized to remove the redundancy of spectral bands, and thus the tri-channel input samples are generated for matching the input of the patch embedding layer in the transformer. Second, two lightweight networks, named spatial enhancement module (SpaEM) and spectral enhancement module (SpeEM), are designed to utilize spectral and spatial information for steering the behavior of the pretrained foundation transformer. Specifically, these two modules can generate spatial and spectral features to modulate the pretrained attention score according to the downstream HSI classification task. Only the two enhancement modules and linear classifier are fine-tuned using limited HSI training samples, which reduces the computational burden and alleviates the overfitting problem. Besides, the idea of relationship distillation based on patches is designed for SS-VFMT, which uses the pretrained knowledge to guide the learning process of the learnable enhancement modules. Finally, the proposed SS-VLFMT uses hash coding to transform HSI image features for facilitating the detection and cognition of unseen classes. The obtained hash codes and text features of class names are aligned to transfer knowledge from seen classes to unseen classes. Based on the above designs, the proposed frameworks can effectively adapt the transformer-based foundation models to HSI's not only traditional but also generalized zero-shot classification tasks.

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The main contributions are summarized as follows.

1) Aiming at exploiting the foundation model's powerful capability at feature extraction, we present an attempt to adapt the transformer-based foundation model to HSI classification and propose a new learning framework named spectral–spatial VFM-based transformer (SS-VFMT). Just by introducing and tuning a small set of learnable parameters related to the spectral–spatial information of HSI, the SS-VFMT can achieve superior performance for HSI classification tasks.
2) To enhance the SS-VFMT's ability to capture local spatial information of HSI, a SpaEM is designed to generate additional input tokens for the frozen transformer. Meanwhile, a SpeEM is designed to enhance the SS-VFMT's ability to capture spectral information of HSI. The generated spectral features are added with query, key, and value tensors within the frozen transformer. The spatial and SpeEMs play roles in modulating the attention scores for better adaption according to the HSI classification task.
3) SS-VFMT-D is proposed by strengthening SS-VFMT with a relationship distillation loss which considers both the interimage and the intraimage patch-level relationship. By focusing on these two kinds of patch-level relationships, the prelearned knowledge from the natural RGB images can serve as a guide for better model learning in HSI classification tasks.
4) Focusing on letting the proposed methods learn to recognize unseen classes, it is the first attempt to propose an SS-VFMT-based framework for HSI generalized zero-shot classification. It aligns SS-VFMT's image features' hash codes and the LFM's class name features to implement knowledge transfer across seen and unseen classes. The proposed SS-VLFMT is an initial effective attempt at generalized ZSL (GZSL) of HSI classification.

The rest of the article is organized as follows. Section II is a brief review of related works. Section III presents the proposed SS-VFMT and SS-VFMT-D for HSI classification. Section IV describes the proposed SS-VLFMT for ZSL in HSI classification. Section V presents the traditional HSI classification experiments. In Section VI, the novel HSI zero-shot classification experiments are conducted, which are exploratory in nature. The conclusion is briefly summarized in Section VII.

II. Related Works

A. CNN-Based Methods for HSI Classification

With the popularity of CNN in deep learning, many CNN-based classification methods have been proposed for HSI and achieved good performance. Hu et al. [41] first utilized 1-D convolution layers to design a five-layer deep model. This model took each pixel vector as an input and mainly extracted the spectral information for HSI classification. Considering that HSIs contain both spatial and spectral information, many spectral–spatial feature extraction methods were proposed using 2-D CNN, 3-D CNN, and a combination of these models. For instance, Zhong et al. [42] introduced an end-to-end spectral–spatial 3-D CNN which utilized the residual connection operations for HSI classification. In [43], the residual unit architecture was integrated into a pyramidal structure to enhance the CNN's classification performance. Li et al. [44] utilized geometric algebra to reformulate and improve the convolution operations, leading a series of novel CNN models which can preserve more structural information among multiple channels. Jia and Lu [45] designed an additional loss to constrain the spatial distribution of the CNN's predicted results, which could make the classification maps more reasonable and provide better classification accuracies. In addition, researchers have explored the combination of CNN with other machine learning techniques, including transfer learning, ensemble learning, and few-shot learning, to enhance CNN's performance in different conditions. In [46], the pixel-pair feature was designed to increase the number of training samples. The ensemble learning strategy was utilized to alleviate the overfitting problem and enhance the stability of CNN-based classifiers. Yu et al. [47] introduced the paradigm of prototype learning to deal with few-shot HSI classification, which is training sample-efficient. Besides, some spatial feature extraction methods were also incorporated into CNNs for better HSI classification performance, such as attribute profiles [48], Gabor filtering [49], and Haar wavelets [50].

B. Transformer-Based Methods for HSI Classification

Transformer-based models have been widely explored in the NLP domain for many years [51]. Thanks to the self-attention mechanism, a transformer can capture long-range dependencies among input tokens. The success of transformers in NLP also attracted the attention of CV researchers and the vision transformer has become popular for various vision tasks, including HSI analysis [52], [53]. For HSI classification, He et al. [54] proposed HSI-BERT which treated every pixel inside a given cube sample as a token for transformer to capture global context. It was the first attempt to use a transformer-based model for HSI classification and achieved competitive accuracies. Considering the long-range dependencies of the spectral bands, Hong et al. [55] proposed SpectralFormer which used a pure transformer to deal with spectral features. Tang et al. [56] designed a double-attention transformer encoder to capture the spatial and spectral features of HSIs, respectively. These features were fused for better discriminative ability. Besides, transformers have been explored to combine with CNNs for feature extraction, which can take advantage of both models. For example, Sun et al. [57] designed a network that contained 2-D and 3-D convolution layers to process the input HSI images first, and then the resulting features were fed into a Gaussian-weighted feature tokenizer to generate input tokens for transformer blocks. To extract the multiscale spectral–spatial features in HSI, Wu et al. [58] proposed an enhanced transformer using multiscale CNNs to generate spectral–spatial tokens with hash-based positional embeddings. Experimental results showed that using multiscale CNN features was beneficial for strengthening the classification ability in HSI.

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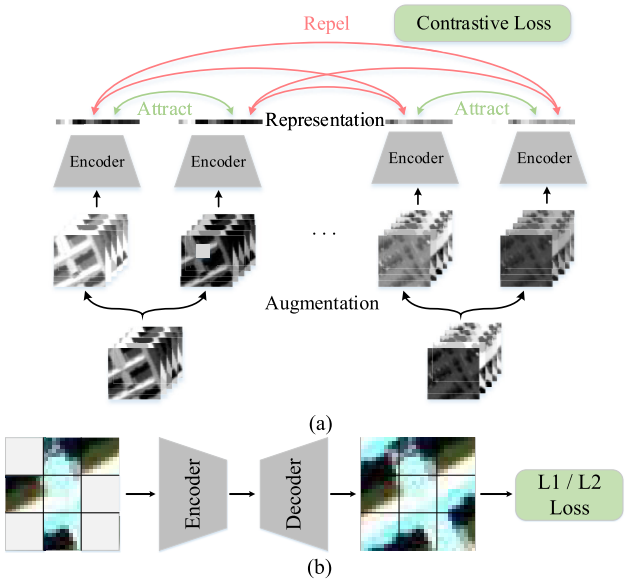


Fig. 1. Brief illustration of (a) CL paradigm and (b) MIM paradigm.

C. Transformer-Based Foundation Models

The transformer-based foundation models in computer vision usually utilized self-supervised learning techniques to pretrain a backbone like ViT-B/16 on large-scale datasets like ImageNet-21K. These foundation models mainly use two kinds of self-supervised learning objectives: contrastive learning (CL) and masked image modeling (MIM). As shown in Fig. 1(a), CL learns general representations by maximizing the similarity between augmented views of the same sample and minimizing the similarity between augmented views of different samples [59], [60], [61]. It is acknowledged that CL depends on the careful selection of augmentations to learn invariances among images [62]. Motivated by BERT for masked language modeling [63], MIM aims to reconstruct the original image [64], or features [65] from the masked input image, as Fig. 1(b) shows. The MIM brings the inductive bias in modeling locality to the transformer model and thus makes the model to be more sample-efficient. However, the learned representations are much less class-discriminative, which limits the performance of downstream discriminant tasks [66]. Therefore, combining the advantages of the above two paradigms effectively is also widely explored for designing transformer-based foundation models. For example, Zhou et al. [67] proposed image BERT pretraining with online tokenizer (iBOT) to integrate MIM and the contrastive self-distillation scheme into vision transforms' pretraining. As an extension of iBOT, the DINO v2 utilized larger datasets and more tricks to obtain models that had more powerful performance in various tasks.

A lot of transformer-based LFMs have emerged in neural language processing at an early time [29]. For example, T5 is a unified pretraining framework proposed by Google, which adopts an encoder–decoder transformer structure [68]. And it regards every text processing problem as a "text-to-text" problem. Due to its openness and validity, LLaMA series models have received a lot of attention from the research and industrial communities since their release [69]. They have achieved excellent performance in all aspects of open benchmarking. Other models like GPT have also shown great success for various tasks [70], [71].

Besides, with the introduction of CLIP and its better generalization ability, the vision-language model gradually gained wide attention [72], [73]. The multimodal foundation model improved performance on various vision and language tasks [74]. Moreover, the vision-language models have also been explored in the remote sensing community [75], [76]. For example, RSGPT is a pretrained model designed for remote sensing visual-linguistic tasks [77]. It consists of an image encoder, a Q-Former for perceptual commands, a fully connected layer, and a large language model LLM, which allows for more diverse tasks such as remote sensing image caption generation and visual question and answer tasks.

III. Proposed SS-VFMT

In this section, we propose an end-to-end spectral–spatial learning framework (SS-VFMT) to apply large-scale pretrained vision transformer-based foundation models for HSI classification tasks, especially when the number of training samples is limited. SS-VFMT only extends the input space of the transformer by introducing a small set of learnable parameters while keeping the backbone frozen during tuning. The designed framework not only saves computational resources but also alleviates the overfitting problem. The overall framework is illustrated in Fig. 2.

It can be seen that the SS-VFMT mainly consists of three components: a pretrained vision transformer that consists of a patch embedding layer and several transformer blocks, a SpaEM, and a SpeEM. In order to utilize the pretrained vision transformer, SS-VFMT utilizes PCA to remove the redundancy of spectral bands, and thus the tri-channel input samples are generated. Subsequently, these samples are spatially resized to match the input of the patch embedding layer in the vision transformer. In the SpaEM, a designed shallow CNN is used to extract features of the original HSI samples and generate additional tokens as a supplement to PCA-based tokens. The PCA-based tokens, together with these additional spatial adaption tokens, are both fed into the transformer blocks for feature extraction. Meanwhile, the SpeEM generates average spectral features for each spatially partitioned patch. Then, these spectral features are added to the corresponding tokens within the transformer blocks, which provide plentiful spectral information into the spatial patches-based model. After spatial–spectral feature extraction by the transformer blocks, a linear layer serves as the classifier that outputs the final classification results.

During tuning on the HSI datasets, only the weights in the SpaEM, the SpeEM, and the linear classifier are optimized. In fact, the SpaEM and the SpeEM play roles in steering the behavior of the frozen vision transformer by modulating the pretrained attention score according to the downstream HSI classification task.

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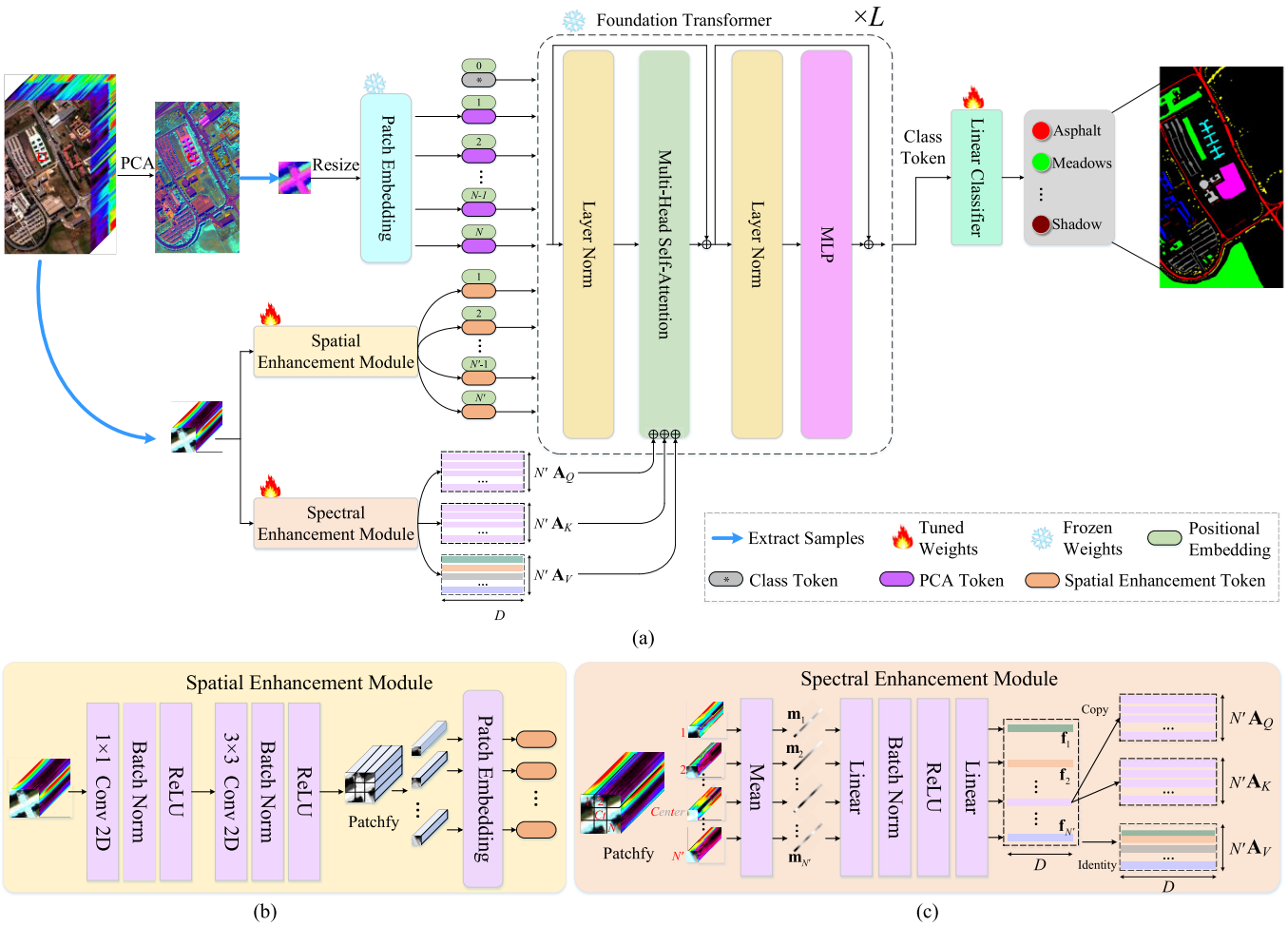


Fig. 2. Illustration for the design details of the proposed SS-VFMT—training flow. For illustrative purpose, a single image flow instead of a batch is shown here. (a) Overall framework (best viewed in color). (b) Details of the designed SpaEM. (c) Details of the designed SpeEM. The large-scale pretrained vision transformer, including a patch embedding layer and several transformer blocks, is frozen during tuning on the HSI datasets. Only the weights in the SpaEM, the SpeEM, and the linear classifier are optimized.

A. Transformer Used in SS-VFMT

The vision transformer is a deep learning model that mainly uses self-attention architecture to process visual inputs. It has been replacing the traditional CNN commonly used in computer vision tasks.

Let's denote the input image as $\mathbf{x} \in \mathbb{R}^{H\times W\times C}$, where H represents the height, W represents the width, and C represents the number of channels (e.g., RGB images have $C = 3$). In a vanilla vision transformer, the input image $\mathbf{x}$ is first partitioned into a sequence of nonoverlapping patches $\{\mathbf{p}_n \in \mathbb{R}^{P^2C}\}_{n=1}^{N}$, where P is the spatial partition size. Each patch $\mathbf{p}_l$ is then linearly projected into a D-dimensional embedding space using a learnable linear transformation $\mathbf{E}$. It is then added with a positional embedding $\mathbf{e}_n \in \mathbb{R}^D$, leading to a patch token

$$\mathbf{z}_n \in \mathbb{R}^D = \mathbf{E}\mathbf{p}_n + \mathbf{e}_n. \tag{1}$$

These patch tokens, together with an additional class token $\mathbf{z}_{\text{cls}}$, compose the input of a standard transformer encoder

$$\mathbf{Z}^0 \in \mathbb{R}^{(N+1)\times D} = [\mathbf{z}_{\text{cls}}; \mathbf{z}_1; \ldots; \mathbf{z}_N]. \tag{2}$$

The transformer encoder usually comprises L blocks, each of which consists of a multihead self-attention (MHSA) layer, multilayer perceptron (MLP) with GELU activation, and layer normalization (LN). The feature extraction process in the lth block can be formulated as

$$\mathbf{H}^l = \text{MHSA}\big(\text{LN}\big(\mathbf{Z}^{l-1}\big)\big) + \mathbf{Z}^{l-1}, \quad l = 1, \ldots, L \tag{3}$$

$$\mathbf{Z}^l = \text{MLP}\big(\text{LN}\big(\mathbf{H}^l\big)\big) + \mathbf{H}^l, \quad l = 1, \ldots, L. \tag{4}$$

After being processed by the transformer encoder, the class token in $\mathbf{Z}^L$ is usually normalized to serve as the representation vector $\mathbf{h}$ for the input image and thus can be used for different tasks

$$\mathbf{h} = LN\big(\mathbf{Z}^L_{\text{cls}}\big). \tag{5}$$

We then use the transformer pretrained on large-scale image datasets like ImageNet-22K as the foundation model. In order to apply a given pretrained vision transformer model, we first transform the HSI image to match the input of the patch embedding layer and use a fully connected layer to obtain the final prediction for the HSI classification task.

Specifically, given an HSI image scene $\mathbf{X}$, the PCA transformation is first used and only the first three principal components are left for generating the tri-channel image. Then, an input sample is extracted from the post-PCA scene

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by using a square neighborhood centered around the given training pixel's location. The resulting tri-channel image is then spatially resized to the desired spatial size to match with the vision transformer model's input. This procedure can be formulated as

$$\bar{\mathbf{X}} = \mathrm{PCA}(\mathbf{X}, 3) \tag{6}$$

$$\mathbf{x} = \mathrm{Resize}\big(\mathrm{Get_sample}\big(\bar{\mathbf{X}}, S, \mathrm{loc}\big)\big) \tag{7}$$

where PCA$(\cdot)$ denotes the operation of PCA, given the number of principal components. Get_sample$(\cdot)$ denotes the operation of extracting a sample from an input scene, given the spatial neighborhood side length S and the training sample's spatial location loc. And Resize$(\cdot)$ means resizing the given image to the desired spatial size using bilinear interpolation.

After that, the obtained input $\mathbf{x} \in \mathbb{R}^{S\times S\times 3}$ is sent to the pretrained vision transformer as described previously. In this case, $N = S^2/P^2$.

As for the output, we add a linear layer to the last transformer block, serving as a classifier. The representation $\mathbf{h}$ for the input PCA image is fed into this layer to obtain the final prediction

$$\hat{\mathbf{y}} \in \mathbb{R}^K = \mathrm{Linear}(\mathbf{h}) \tag{8}$$

where Linear$(\cdot)$ denotes the function of a linear layer and $\hat{\mathbf{y}}$ is the predicted label vector for the input image. K is the number of classes in the HSI dataset.

Given the input sample's label, the cross-entropy loss is used for optimizing the designed model

$$\mathcal{L}_{\mathrm{cls}} = -\sum_{k=1}^{K} y_k \log \hat{y}_k \tag{9}$$

where $\hat{y}_k$ is the kth element of $\hat{\mathbf{y}}$, and y_k is the kth element of the one-hot label vector.

B. SpaEM for HSI Classification

Recent researches show that incorporating convolutions into transformers can enhance their ability to capture local spatial information in HSI classification [78]. Building upon this insight, we propose the SpaEM to model the local spatial contexts of HSIs alongside the original transformer's patch embedding layer. This approach allows us to preserve the original architecture of the vision transformer, while effectively considering the local spatial information of HSI.

As shown in Fig. 2(b), a shallow CNN, which mainly consists of a 1×1 and a 3×3 convolution layers, is first employed to extract features from any given HSI input image $\tilde{\mathbf{x}} \in \mathbb{R}^{S\times S\times B}$, where B denotes the number of bands. The procedure mentioned above can be summarized as

$$\tilde{\mathbf{x}} = \mathrm{get_sample}(\mathbf{X}, S, \mathrm{loc}) \tag{10}$$

$$\mathbf{q} \in \mathbb{R}^{H\times W\times D'} = \mathrm{CNN}\big(\tilde{\mathbf{x}}\big) \tag{11}$$

where D' denotes the hidden dimension for the feature map and CNN$(\cdot)$ denotes the feature extraction process of the shallow CNN.

Then, the feature map $\mathbf{q}$ is spatially partitioned into nonoverlapped patches $\{\tilde{\mathbf{p}}_n \in \mathbb{R}^{P'^2 D'}\}_{n=1}^{N'}$ with spatial size P' and linearly projected into tokens, just as described previously in Section III-A

$$\tilde{\mathbf{z}}_n \in \mathbb{R}^D = \mathbf{E}'\tilde{\mathbf{p}}_n + \tilde{\mathbf{e}}_n \tag{12}$$

$$\mathbf{M}^0 \in \mathbb{R}^{(N'+1)\times D} = [\mathbf{z}_1; \ldots; \mathbf{z}_{N'}] \tag{13}$$

where $\mathbf{E}'$ is the corresponding projection matrix and $\tilde{\mathbf{e}}_n$ is the nth positional embedding and $N' = S^2/P'^2$.

Finally, the obtained tokens are concatenated to the PCA tokens, forming the input of the transformer encoder denoting as $\hat{\mathbf{Z}}^0$

$$\hat{\mathbf{Z}}^0 \in \mathbb{R}^{(N+N'+1)\times D} = [\mathbf{Z}^0 || \mathbf{M}^0] \tag{14}$$

where $||$ denotes the concatenation operation. This operation ensures that every token interacts with all other tokens through the self-attention layers at each layer of the model. The feature extraction process after concatenation can be formulated as

$$\hat{\mathbf{H}}^l = \mathrm{MHSA}\big(\mathrm{LN}\big(\hat{\mathbf{Z}}^{l-1}\big)\big) + \hat{\mathbf{Z}}^{l-1}, \quad l = 1, \ldots, L \tag{15}$$

$$\hat{\mathbf{Z}}^l = \mathrm{MLP}\big(\mathrm{LN}\big(\hat{\mathbf{H}}^l\big)\big) + \hat{\mathbf{H}}^l, \quad l = 1, \ldots, L \tag{16}$$

$$\mathbf{h} = \mathrm{LN}\big(\hat{\mathbf{Z}}^L_{\mathrm{cls}}\big). \tag{17}$$

Discussion: The spatial enhancement tokens endow the model with the ability to capture features associated with HSI classification tasks. In addition to strengthening the ability to capture local information, the spatial enhancement tokens also reweight the pretrained attention score, leading to better adaption for HSI classification tasks. For better understanding the role of these spatial enhancement tokens, we explore the attention score in the MHSA

$$\tilde{w}_{ij} \propto \frac{\exp\left(\frac{\mathbf{q}_i^T \mathbf{k}_j}{\sqrt{D}}\right)}{\sum_{j=1}^{N} \exp\left(\frac{\mathbf{q}_i^T \mathbf{k}_j}{\sqrt{D}}\right) + \sum_{j=1}^{N'} \exp\left(\frac{\mathbf{q}_i^T \mathbf{k}_j}{\sqrt{D}}\right)}. \tag{18}$$

Dividing the numerator and denominator by $\sum_{j=1}^{N} \exp(\mathbf{q}_i^T \mathbf{k}_j/\sqrt{D})$, we have

$$\tilde{w}_{ij} = \frac{w_{ij}}{1 + \beta w_{ij}} \tag{19}$$

where $\beta = \sum_{j=1}^{N'} \exp(\mathbf{q}_i^T \mathbf{k}_j/\sqrt{d}) / \sum_{j=1}^{N} \exp(\mathbf{q}_i^T \mathbf{k}_j/\sqrt{d})$. It can be seen that β is a coefficient that is derived from the enhancement tokens and can be optimized for better adapting to the downstream HSI classification task. We can infer that the task-specific spatial enhancement tokens play a role in modulating the attention weights, where their names are derived.

C. SpeEM for HSI Classification

The vision transformer is mainly designed to process the spatial relationship between patches. To fully utilize the spectral features in HSI images, we designed the spectral-enhanced module. Fig. 2(c) shows the detailed architecture of the designed SpeEM. This module first obtains each patch's mean spectral features. These features are then projected to be aligned with the embedding dimension in transformer blocks. After that, they will be added with query, key, and value tensors in the MHSA layers, respectively.

Specifically, following the way of spatial partition in the SpaEM, the input HSI image $\mathbf{X}$ is divided into N' patches

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$\{\hat{\mathbf{p}}_n \in \mathbb{R}^{P'^2 \times B}\}_{n=1}^{N'}$. Then, the mean spectral feature $\mathbf{m}_i \in \mathbb{R}^B$ for each patch is computed by averaging the pixels for robust spectral information. This process can be formulated by

$$\mathbf{m}_i = \frac{1}{P'^2} \sum_{j=1}^{P'^2} \hat{\mathbf{p}}_i^j, \quad i = 1, \ldots, N' \tag{20}$$

where j denotes the jth pixels in this patch.

After that, the mean spectral feature $\mathbf{m}_i$ for each patch is processed by an MLP for further feature reduction and dimension matching, leading to feature $\mathbf{f}_i \in \mathbb{R}^D$

$$\mathbf{f}_i = \text{MLP}(\mathbf{m}_i), \quad i = 1, \ldots, N' \tag{21}$$

where MLP denotes a multilayer perceptron that deals with spectral features. Each patch finally has a corresponding spectral feature vector $\mathbf{f}_i$, just like Fig. 2(c) shows. These spectral features $\{\mathbf{f}_i\}_{i=1}^{N'}$ are rearranged into spectral feature matrix $\mathbf{F}$

$$\mathbf{F} \in \mathrm{R}^{N' \times D} = [\mathbf{f}_1; \ldots; \mathbf{f}_{N'}] \tag{22}$$

where each row represents a specific patch's spectral feature vector.

Subsequently, the obtained spectral features $\mathbf{F}$ are utilized to form three tensors, named $\mathbf{A}_Q$, $\mathbf{A}_K$, and $\mathbf{A}_V$. To get the $\mathbf{A}_Q$ and $\mathbf{A}_K$, we make N' copies of the central patch's spectral feature. The procedure can be formulated as

$$\mathbf{A}_Q = \mathbf{A}_K = [\mathbf{f}_{(N'+1)/2}; \mathbf{f}_{(N'+1)/2}; \ldots; \mathbf{f}_{(N'+1)/2}] \in \mathbb{R}^{N' \times D}. \tag{23}$$

That is, each row in the $\mathbf{A}_Q$ and $\mathbf{A}_K$ matrices is the central patch's corresponding spectral feature vector $\mathbf{f}_{(N'+1)/2}$. $\mathbf{A}_V$ is directly derived from $\mathbf{F}$, as they are exactly the same

$$\mathbf{A}_V = \mathbf{F}. \tag{24}$$

The reason why the center patch's feature is used to generate $\mathbf{A}_Q$ and $\mathbf{A}_K$ will be explained later. These obtained tensors are going to be added with query, key, and value tensors in the MHSA, respectively.

For the MHSA in the lth block in the transformer, three tensors named query $\mathbf{Q}^l$, key $\mathbf{K}^l$, and value $\mathbf{V}^l$ are first generated

$$\mathbf{Q}^l = \text{LN}(\hat{\mathbf{Z}}^{l-1})\mathbf{W}^{Q,l} \tag{25}$$
$$\mathbf{K}^l = \text{LN}(\hat{\mathbf{Z}}^{l-1})\mathbf{W}^{K,l} \tag{26}$$
$$\mathbf{V}^l = \text{LN}(\hat{\mathbf{Z}}^{l-1})\mathbf{W}^{V,l} \tag{27}$$

where $\mathbf{W}^{Q,l}$, $\mathbf{W}^{K,l}$, and $\mathbf{W}^{V,l}$ denote the query, key, and value projection matrices in the MHSA, respectively. Then, the computation of MHSA at the lth block can be represented by

$$\begin{aligned} \mathbf{r}^l &= \text{Attention}(\mathbf{Q}^l, \mathbf{K}^l, \mathbf{V}^l) \\ &= \text{Softmax}\left(\frac{\mathbf{Q}^l \mathbf{K}^{l^T}}{\sqrt{D}}\right)\mathbf{V}^l \end{aligned} \tag{28}$$

$$\mathbf{O}^l = \mathbf{r}^l \mathbf{W}^{lp} \tag{29}$$

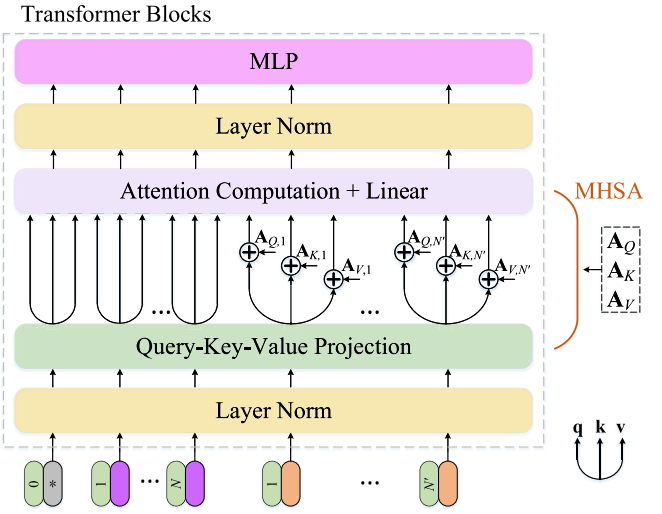


Fig. 3. Procedure of adding the spectral features into the MHSA layer.

where $\mathbf{W}^{lp}$ means linear projection and $\mathbf{O}^l$ represents the output of MHSA. For illustration purpose, the multihead operation is omitted here.

Thus, the obtained spectral enhancement features are added to the original query, key, and value tensors

$$\mathbf{Q}^l = \mathbf{Q}^l + [\mathbf{0}||\mathbf{A}_Q] \tag{30}$$
$$\mathbf{K}^l = \mathbf{K}^l + [\mathbf{0}||\mathbf{A}_K] \tag{31}$$
$$\mathbf{V}^l = \mathbf{V}^l + [\mathbf{0}||\mathbf{A}_V] \tag{32}$$

where $\mathbf{0} \in \mathbb{R}^{(N+1)\times D}$ denotes the zeros matrix, and it is concatenated with spectral enhancement features. Fig. 3 illustrates the procedure for using these adaptive features in the MHSA. From (30)–(32), one can see that only the spatial adaptive tokens are added with spectral enhancement features. The updated query, key, and value tensors are then used to complete the computation in the MHSA, just as described above.

Discussion: For better understanding the role of these spectral enhancement features, the computation process of attention is also explored in Section III-B

$$\begin{aligned} \tilde{w}_{ij} &\propto \exp\left(\frac{\mathbf{q}_i^T(\mathbf{k}_j + \mathbf{A}_{K,j})}{\sqrt{D}}\right) \\ &= \exp\left(\frac{\mathbf{q}_i^T \mathbf{k}_j}{\sqrt{D}}\right) \exp\left(\frac{\mathbf{q}_i^T \mathbf{A}_{K,j}}{\sqrt{D}}\right) \\ &= w_{ij} \cdot \alpha_{ij}. \end{aligned} \tag{33}$$

From the above (33), it can be seen that the SpeEM provides spectral features that can reweight the original vision transformer's attention scores. As the query and key are nearly symmetric, we set $\mathbf{A}_Q$ and $\mathbf{A}_K$ the same. Note that the input HSI image's label is mainly determined by the center pixel's label and the spectral information is mainly needed here for enhancement. Thus, we hope the reweight factor α_{ij} is mainly determined by the center area's spectral information, and tries to exclude the influence of the pixels that have little correlation with the center pixel. That is why we use the center patch's feature to generate $\mathbf{A}_Q$ and $\mathbf{A}_K$.

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D. Relationship Distillation for SS-VFMT

Generally speaking, both the PCA tokens and spatial enhancement tokens mainly capture the spatial features of HSI in the given pretrained transformer. The PCA tokens are prelearned and fixed during the training of SS-VFMT, while the SpaEMs are initialized and optimized for better adaption. Intuitively, the idea of "knowledge distillation" can be applied so that the PCA tokens can serve as guidance for the learning procedure of the SpaEM. Therefore, in this section, we design a relationship distillation method focusing on the relationships among the PCA tokens and spatial enhancement tokens.

Consider a batch of samples with a size of Δ, where $\mathbf{T}_p \in \mathbb{R}^{\Delta\times N\times D}$ and $\mathbf{T}_{ss} \in \mathbb{R}^{\Delta\times N'\times D}$ represent the transformer encoder's output tokens for the PCA-branch and the spatial–spectral adaption branch, respectively. Here, we resize the $\mathbf{T}_p$ so that it has the same shape as $\mathbf{T}_{ss}$. That is, $\mathbf{T}_p \in \mathbb{R}^{\Delta\times N'\times D}$.

To model the patch-level relationships among a batch of images, these tokens are first reshaped and normalized along the last dimension, and then, the similarity map between these tokens is computed

$$\mathbf{S}(\Omega(\mathbf{T}_p)) = \Omega(\mathbf{T}_p)\Omega(\mathbf{T}_p)^{\mathrm{T}} \tag{34}$$

where Ω means reshaping any given tensor: $\mathbb{R}^{D_1\times D_2\times D_3} \to \mathbb{R}^{D_1 D_2\times D_3}$. Similarly, we can obtain the similarity map of $\mathbf{T}_{ss}$. Thus, we design a distillation loss that uses the PCA tokens to facilitate the training of the enhancement module

$$\mathcal{L}_d = \left\|\mathbf{S}\left(\Omega\left(\mathbf{T}_p\right)\right) - \mathbf{S}(\Omega(\mathbf{T}_{ss}))\right\|_F^2. \tag{35}$$

From the above equation, it can be seen that the relationship among patches of a batch is taken into consideration. However, the computation burden is too heavy to afford. Therefore, the similarity map is decoupled into an interimage similarity map, which mainly captions the similarity relationship of images, and an intraimage similarity map which mainly captions the patch-level similarity within an image. Thus, we have the intraimage patch-level distillation loss

$$\mathcal{L}_{\text{intra}} = \frac{1}{\Delta}\sum_{i=0}^{\Delta}\left\|\mathbf{S}\left(\mathbf{T}_p[i,:,:]\right) - \mathbf{S}(\mathbf{T}_{ss}[i,:,:])\right\|_F^2. \tag{36}$$

Similarly, the interimage patch-level distillation loss is computed by

$$\mathcal{L}_{\text{iner}} = \frac{1}{N'}\sum_{j=0}^{N'}\left\|\mathbf{S}\left(\mathbf{T}_p[:,j,:]\right) - \mathbf{S}(\mathbf{T}_{ss}[:,j,:])\right\|_F^2. \tag{37}$$

Fig. 4 illustrates the different relationship schemes in this study.

The total loss function for the SS-VFMT-D is formulated as

$$\mathcal{L}_{\text{total}} = \mathcal{L} + \alpha(\mathcal{L}_{\text{intra}} + \mathcal{L}_{\text{iner}}) \tag{38}$$

where α is a coefficient that is used to balance the classification loss and the distillation loss.

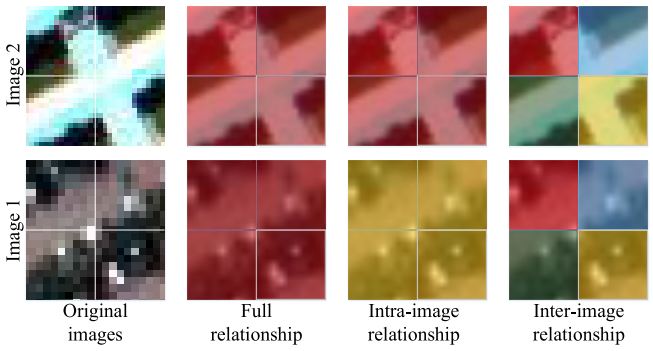


Fig. 4. Illustration of relationship distillation. Similarity relationships are computed for every set of patches filled with the same color.

IV. Proposed SS-VLFMT

Though the proposed SS-VFMT-based methods can utilize the pretrained knowledge to benefit the HSI classification task, they are still limited to a closed-set environment. When the test set contains unseen classes that are not labeled in the training set directly, the proposed method fails to classify. However, the existing vision-LFMs have shown great potential for ZSL ability. Focusing on letting the proposed methods learn to recognize unseen classes with the help of vision-LFMs, we propose an SS-VFMT-based framework. It utilizes both pretrained vision and LFMs to benefit HSI zero-shot classification.

To start with, we simply illustrate the ZSL problem for HSI classification. Given an HSI dataset D, which consists of a training set D_{tr} for model training and a testing set D_{te} for testing, we have $D_{\text{tr}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N} \in X_{\text{tr}} \times Y_{\text{tr}}$ and $D_{\text{tr}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{M} \in X_{\text{te}} \times Y_{\text{te}}$, where $\mathbf{x}_i$ represents an HSI image sample and y_i denotes the corresponding label. We define the classes in the training set as seen classes, denoted as Y_{S}. Thus, we have $Y_{\text{tr}} = Y_{\text{S}}$. The classes which are not in the training set but appear in the test set as unseen classes Y_{U}. For ZSL, we have $Y_{\text{te}} = Y_{\text{U}}$. That is, the samples in the test set all come from the unseen classes, which are not contained in the training set. For GZSL, we have $Y_{\text{te}} = Y_{\text{S}} \cup Y_{\text{U}}$. That is, the test set contains samples belonging to both seen classes in the training set and unseen classes. GZSL is more challenging but realistic in HSI classification. Therefore, this study mainly focuses on GZSL. For the sake of convenience, all the ZSL/classification in the following text refers to GZSL/classification.

The proposed GZSL framework (SS-VLFMT) for HSI classification is shown in Fig. 5. Generally speaking, the whole framework consists of three parts: an SS-VFMT-based image encoder, an LFMT-based text encoder, and a vision-language alignment. Given an input HSI sample, its image input is first processed by the proposed SS-VFMT image encoder for spectral–spatial feature extraction. The extracted feature (i.e., class token) is then fed into two networks, i.e., sign head and amplitude head. The element-wise multiplication of the obtained sign feature and amplitude feature are considered as the final image feature for the given HSI image input, and the sign feature is then processed using hash coding. Thus, an image feature and a hash code feature are obtained for each

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image input after being fed into the vision encoder. The image feature is then used for supervised CL. Meanwhile, the class names are fed into a well-trained text encoder, e.g., CLIP text encoder, to get text features for all classes. Finally, the hash code feature and the text features are used to align the image with the corresponding class name.

A. Image Branch of SS-VLFMT

The image branch mainly consists of an SS-VFMT and two MLP networks, named sign head and amplitude head, respectively. Given a sample, it contains a 3-D HSI image $\tilde{\mathbf{x}}_i$, the corresponding PCA image $\mathbf{x}_i$, and label y_i. The proposed SS-VFMT is used for spectral–spatial feature extraction. The obtained class token is denoted as $\mathbf{z}_i$

$$\mathbf{z}_i = \mathcal{F}(\mathbf{x}_i, \tilde{\mathbf{x}}_i) \tag{39}$$

where $\mathcal{F}$ represents the mapping function of SS-VFMT.

To support the recognition of both seen and unseen classes in an HSI dataset, we use hash coding to transform the embedding space to a hash space where the decision boundary can easily accommodate both seen and unseen classes. This method assigns a unique hash code to each class and frees the model from in-class prediction. For a given vector $\mathbf{w} = (w_1, w_2, \ldots, w_l, \ldots, w_L)$, where L is the vector's length, the hash code of $\mathbf{w}$ can be obtained using

$$\text{Hash}(\mathbf{w}) = \left(w_1^*, w_2^*, \ldots, w_l^*, \ldots, w_L^*\right) \tag{40}$$

$$w_l^* = \begin{cases} 1, & w_l \geq 0 \\ -1, & w_l < 0. \end{cases} \tag{41}$$

Through hash coding, we cannot only distinguish unseen classes' samples from seen classes but also cluster them.

In an ideal scenario, each dimension of the hash codes represents a distinct high-level semantic concept, such as texture or water absorption features in HSI. However, due to the complexity of land-cover features in HSI, there exists the issue that the hash coding is sensitive to intraclass variance, which often leads to the prediction of many nonexistent classes. To mitigate this problem, we separate the feature embedding into two distinct sign and magnitude components through sign head $\mathcal{H}_S$ and amplitude head $\mathcal{H}_A$ in parallel. It is hoped that the sign component is class level and the magnitude component represents the intraclass variance in HSI.

The processed feature $\mathbf{z}_i$ is fed into the two heads, and the corresponding sign feature $\mathbf{s}_i$ and amplitude feature $\mathbf{a}_i$ are obtained

$$\mathbf{s}_i = \mathcal{H}_S(\mathbf{z}_i) = \text{Tanh}(\text{MLP}_S(\mathbf{z}_i)) \tag{42}$$

$$\mathbf{a}_i = \mathcal{H}_A(\mathbf{z}_i) = \text{MLP}_A(\mathbf{z}_i)/\text{Tanh}(\text{MLP}_A(\mathbf{z}_i)) \tag{43}$$

where MLP_S and MLP_A are two multilayer perceptrons and $\text{Tanh}(\cdot)$ means the hyperbolic tangent function

$$\text{Tanh}(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}. \tag{44}$$

It can be seen that (42) is a smoothed version of the hash coding function and the sign feature can be approximated as the corresponding hash code. This scheme is adopted mainly for easier gradient propagation.

A regularization term is used to force the sign feature's absolute value to be close to 1 or -1. This can be achieved by using L_1 norm

$$\mathcal{L}_{\text{reg}} = -|\mathbf{s}_i|_1. \tag{45}$$

The final image feature is obtained by element-wise multiplication of $\mathbf{s}_i$ and $\mathbf{a}_i$

$$\mathbf{f}_i = \mathbf{s}_i \cdot \mathbf{a}_i. \tag{46}$$

The supervised CL loss is used here for training, as it does not need the softmax function and is not limited to the seen classes. Let $s(\mathbf{a}, \mathbf{b}) = \mathbf{a}^{\text{T}}\mathbf{b}/(\|\mathbf{a}\|\|\mathbf{b}\|)$ represents the inner product between l_2 normalized vectors $\mathbf{a}$ and $\mathbf{b}$. Then, the supervised contrastive loss function is defined as

$$\mathcal{L}_{\text{scl}} = \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{e^{s(\mathbf{f}_i, \mathbf{f}_p)}}{\sum_{k=1}^{Q} \mathbb{1}_{[k \neq i]} e^{s(\mathbf{f}_i, \mathbf{f}_k)}} \tag{47}$$

where $\mathbb{1}_{[\cdot]} \in \{0, 1\}$ is an indicator function returning 1 if and only if the condition in $[\cdot]$ is satisfied. $P(i)$ means the indices of positive samples in the current mini-batch, and Q is the batch-size.

B. Text Branch of SS-VLFMT

The text encoder can be a pretrained transformer, such as BERT and CLIP text encoder. Without generality, the CLIP text encoder is used here to associate each land-cover's name to a specific feature vector.

For each class, the template "A photograph of a ⟨class name⟩" is used to construct the input of the text encoder, e.g., "A photograph of a building." Then, the formed class text inputs are fed into the text encoder, and the text features for all the seen and unseen classes are obtained

$$\mathbf{t}_i = \text{Text}\left(\text{tp}_i\right), \quad i \in Y_{\text{te}} \tag{48}$$

where tp_i means the template of class i, $\mathbf{t}_i$ means the corresponding text feature, and $\text{Text}(\cdot)$ represents the text encoder's mapping function.

C. Image–Text Alignment of SS-VLFMT

The hashing coding scheme assigns a unique hash code for each class. To align each HSI image's hash code with its class name, the alignment module is designed here.

As the pretrained text encoder has a fixed output dimension, the sign feature $\mathbf{s}_i$ is first fed into a linear layer to match the text encoder's output dimension

$$\mathbf{v}_i = \text{Linear}(\mathbf{s}_i) \tag{49}$$

and then the logits $\mathbf{p}_i$ can be obtained using $\mathbf{v}_i$ and the text feature

$$\mathbf{p}_i = \mathbf{T}\mathbf{v}_\mathbf{i}^{\mathbf{T}} \tag{50}$$

where $\mathbf{T} \in \mathbb{R}^{|Y_{\text{te}}| \times D_l}$ is the class text feature matrix, whose ith row is the ith class's text feature. $\mathbf{D}_l$ denotes the text encoder's output dimension.

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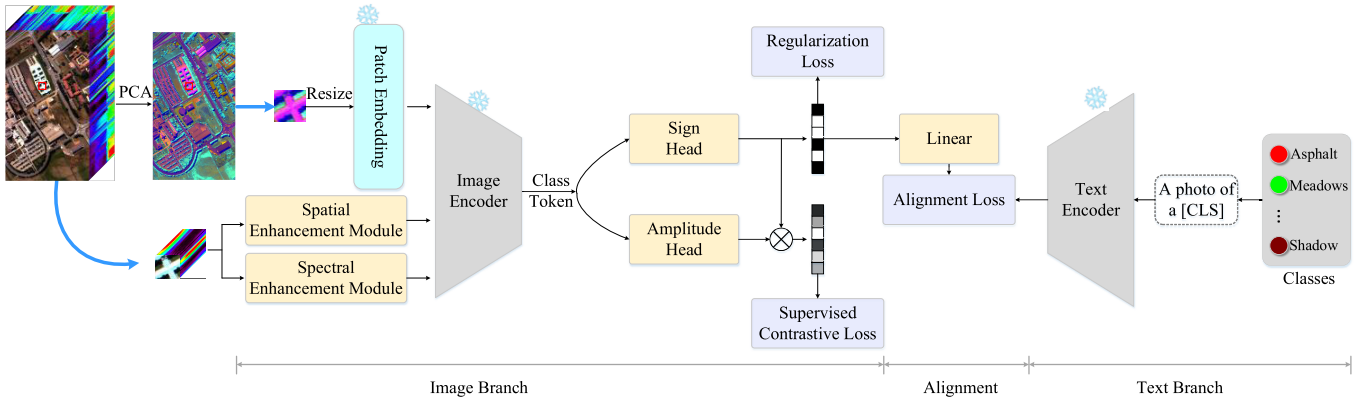


Fig. 5. Illustration of the proposed SS-VLFMT—training flow. For illustrative purpose, a single image flow instead of a batch is shown here. The framework mainly contains three parts: 1) an SS-VFMT-based image encoder, which gets hash coding-based image feature and corresponding hash code for unseen classes cognition; 2) an LFMT-based text encoder, which transforms class names into features vector; and 3) a vision-language alignment part, which aligns image hash code with corresponding class name's text feature.

The alignment can be achieved by using common cross-entropy loss

$$\mathcal{L}_{\text{it}} = -\sum_{c \in Y_{\text{te}}} y_i^c \log p_i^c \tag{51}$$

where $\mathbf{y}_i$ is the one-hot label of the ith training samples and c is the index of each class.

D. Training and Inference of SS-VLFMT

During training, the parameters in the pretrained vision transformer in the SS-VFMT and text transformer in the text encoder are frozen, while the rest parameters are trained. The total training loss function for HSI zero-shot classification is the weighted sum of CL loss, regularization loss, and alignment loss

$$\mathcal{L}_{\text{zsl}} = \mathcal{L}_{\text{scl}} + \beta \mathcal{L}_{\text{reg}} + \gamma \mathcal{L}_{\text{it}}. \tag{52}$$

After training well, each sample in the training set will get a hash code. We will establish a hash code book B which associates each class in the training set with its hash code. Specifically, for the given ith HSI training sample, we first obtain the corresponding sign feature $\mathbf{s}_i$. The obtained sign feature $\mathbf{s}_i$ is then processed by hash coding [i.e., (40) and (41)], and the hash code $\mathbf{h}_i$ is obtained

$$\mathbf{h}_i = \text{Hash}(\mathbf{s}_i). \tag{53}$$

After hash coding, sample features which are slightly different due to noise or variation are degraded to unique hash codes, whose elements are **binary**. Considering that the network is well trained, after hash coding, each class will get a corresponding unique hash-coded vector as a unique sign of the class. These hash-coded vectors are collected to establish a hash code book B. As shown in Fig. 6, each of Classes 3 and 5 contains 3 training samples. The samples in the same class have slightly different sign features. However, after hash coding, they all correspond to a unique vector whose elements are 1 or -1.

In the inference stage, given a test sample, including a 3-D HSI image $\tilde{\mathbf{x}}_j$ and the corresponding PCA image $\mathbf{x}_j$, we first obtain the corresponding hash code $\mathbf{h}_j$ and predict logits $\mathbf{p}_j$.

Then, we need to judge whether the obtained hash code $\mathbf{h}_j$ is in the hash code book or not. The sample can be considered from seen classes if the hash code is exactly the same as the one in the codebook. As shown in Fig. 6, the second test sample's hash code is exactly the same as the hash code of Class 5 in the codebook, and it most likely belongs to Class 5. On the contrary, if the corresponding hash code of the test sample cannot be found in the codebook, it can be assumed that this sample belongs to the unknown class. As shown in Fig. 6, the hash code of the first and third test samples cannot be found in the codebook with the corresponding class hash code. Besides, the hash codes of the two samples are not consistent, so it is judged that they belong to different clusters.

Therefore, the use of hash coding allows the model to not only classify samples of seen classes but also to cluster the samples that the model thinks from unseen classes. And we do not need to know how many clusters we should use. If we set the code's length to be eight (i.e., $L = 8$), there will be 2^L distinct classes that can be represented potentially. In other words, the hash-based decision space has a maximum number of categories 2^L, which is usually enough to cover the land-covers in a given HSI scene.

The final predict class $\check{c}_j$ of the given HSI sample is then obtained by

$$\check{c}_j = \begin{cases} \underset{c \in Y_{\text{S}}}{\text{argmax}}\, p_j^c, & \text{if } \mathbf{h}_j \in B \\ \underset{c \in Y_{\text{U}}}{\text{argmax}}\, p_j^c, & \text{else.} \end{cases} \tag{54}$$

It can be seen that the hash coding helps discover new clusters, and the image–text alignment plays a role in labeling these clusters into unseen classes.

V. Experiments of Traditional HSI Classification

This section of the experiments is mainly about using the proposed SS-VFMT and SS-VFMT-D for traditional HSI classification, which are comprehensive and detailed as follows.

A. Datasets Description

In the experiments, the performance of the proposed methods is evaluated on three widely used HSI datasets, including

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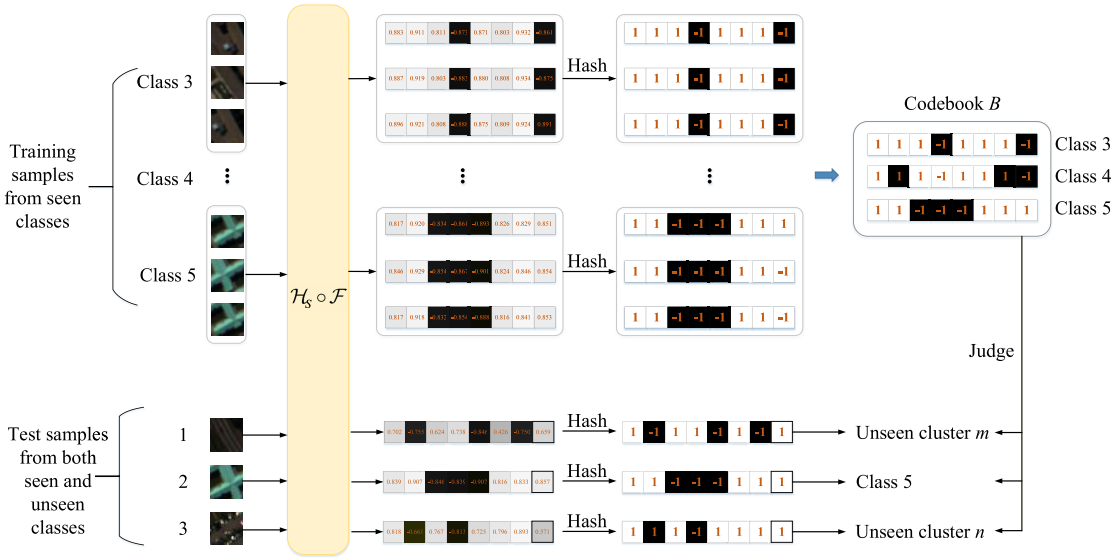


Fig. 6. Illustration of using hash codes for new class discovery. After hash coding, sample features which belong to the same cluster but are slightly different due to noise or variation are degraded to unique hash codes, whose elements are **binary**. The feature space has been transformed into a structural hash space where the decision boundaries are inherent—assigning each class a unique hash code. During the test, samples sharing the same hash code are treated as a cluster.

TABLE I

LAND COVER CLASSES AND NUMBERS OF SAMPLES FOR THE INDIAN PINES DATASET

No.	Class Name	Training samples	Test samples	Total samples
1	Alfalfa	20	26	46
2	Corn-notill	20	1408	1428
3	Corn-mintill	20	810	830
4	Corn	20	217	237
5	Grass-pasture	20	463	483
6	Grass-trees	20	710	730
7	Grass-pasture-mowed	20	8	28
8	Hay-windrowed	20	458	478
9	Oats	15	5	20
10	Soybean-notill	20	952	972
11	Soybean-mintill	20	2435	2455
12	Soybean-clean	20	573	593
13	Wheat	20	185	205
14	Woods	20	1245	1265
15	Buildings-Grass-Trees	20	366	386
16	Stone-Steel-Towers	20	73	93
	Total	315	9934	10249

the Indian Pines dataset, the Pavia University dataset, and the Houston dataset.

1) Indian Pines: This dataset primarily focuses on the agricultural fields in Northwestern Indiana, USA. It was captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor in June 1992. It consists of 145 × 145 pixels, and the spatial resolution is 20 m. Besides, this dataset includes 220 spectral bands, with wavelengths ranging from 400 to 2500 nm. For evaluation purposes, 200 bands are preserved after excluding 20 low signal-to-noise ratio (SNR) bands. The dataset contains 10 249 labeled samples and 16 different land-cover types. Fig. 7 shows the false color map and corresponding ground-truth map of the Indian Pines dataset. Furthermore, Table I presents the number of training and test samples per class.

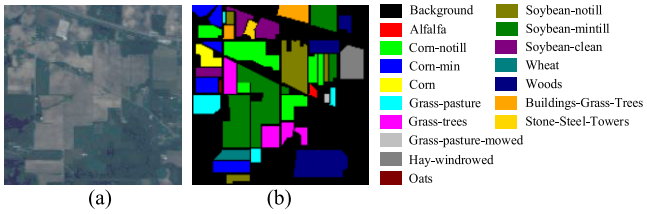


Fig. 7. Indian Pines dataset: (a) false color map and (b) ground-truth.

TABLE II

LAND COVER CLASSES AND NUMBERS OF SAMPLES FOR THE PAVIA UNIVERSITY DATASET

No.	Class Name	Training samples	Test samples	Total samples
1	Asphalt	20	6611	6631
2	Meadows	20	18629	18649
3	Gravel	20	2079	2099
4	Trees	20	3044	3064
5	Metal sheets	20	1325	1345
6	Bare soil	20	5009	5029
7	Bitumen	20	1310	1330
8	Bricks	20	3662	3682
9	Shadow	20	927	947
	Total	180	42596	42776

2) Pavia University: The Reflective Optics System Imaging Spectrometer (ROSIS) sensor captured this dataset over the University of Pavia, Italy. It encompasses 610 × 340 pixels with a spatial resolution of 1.3 × 1.3 m. A total number of 103 spectral bands are recorded in this dataset. The ground truth labeled 42 776 pixels that belong to nine classes. Fig. 8 displays the false color map and corresponding ground-truth map of the dataset, while Table II provides the number of training and test samples per class.

3) Houston: This dataset was captured by the National Center for Airborne Laser Mapping (NCALM) [79]. It describes

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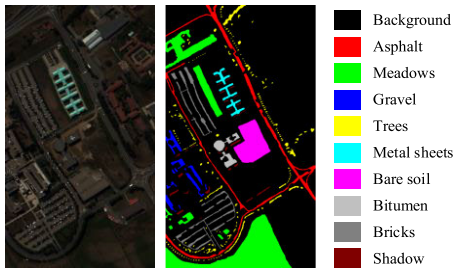


Fig. 8. Pavia University dataset: (a) false color map and (b) ground-truth.

TABLE III
LAND COVER CLASSES AND NUMBERS OF SAMPLES FOR THE HOUSTON DATASET

No.	Class Name	Training samples	Test samples	Total samples
1	Grass-healthy	20	1231	1251
2	Grass-stressed	20	1234	1254
3	Grass-synthetic	20	677	697
4	Tree	20	1224	1244
5	Soil	20	1222	1242
6	Water	20	305	325
7	Residential	20	1248	1268
8	Commercial	20	1224	1244
9	Road	20	1232	1252
10	Highway	20	1207	1227
11	Railway	20	1215	1235
12	Parking-lot-1	20	1213	1233
13	Parking-lot-2	20	449	469
14	Tennis-court	20	4008	4028
15	Running-track	20	640	660
	Total	300	14729	15029

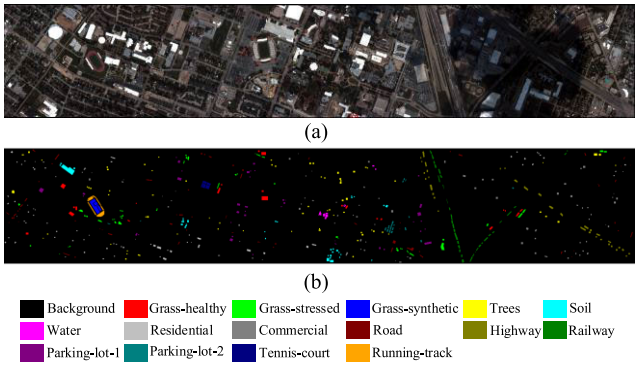


Fig. 9. Houston dataset: (a) false color map and (b) ground-truth.

an urban area surrounding the University of Houston campus in the USA. This dataset was used for the 2013 IEEE GRSS Data Fusion Contest and contains 349 × 1905 pixels, with a spatial resolution of 2.5 × 2.5 m. The dataset also encompasses 144 bands with a wavelength range of 380–1050 nm. It contains 15 029 labeled pixels belonging to 15 different land-cover types in this dataset. Fig. 9 depicts the false color map and corresponding ground-truth map, while Table III provides the number of training and test samples per class.

B. Experimental Setup

The effectiveness of the proposed SS-MTr-based methods is demonstrated through comparison with various HSI classification methods as follows.

1) *EMP-SVM [9]:* It is a traditional method that utilizes EMP for spatial feature extraction and SVM for final classification. It serves as a benchmark for comparison with deep learning-based methods.
2) *CNN:* It is a vanilla CNN with four convolutional layers, representing a basic spectral–spatial model for HSI classification. And we use it to highlight the imitations of a vanilla transformer.
3) *SSRN [42]:* SSRN is an end-to-end CNN-based residual network that leverages stacked 3-D convolutional layers to extract discriminative spectral–spatial features for HSI classification.
4) *DBDA [80]:* DBDA is an improved CNN that uses a double-branch dual-attention mechanism for more discriminative feature extraction. This method uses an attention mechanism that differs from transformer-based methods, which use self-attention mechanisms.
5) *FDSSC [81]:* This method designs a fast dense spectral–spatial convolution network that extracts spectral and spatial features separately using different convolutional kernel sizes. It is specifically an improved CNN that can achieve competitive accuracies.
6) *SS-Tr:* SS-Tr is a vanilla transformer with four transformer blocks, which uses a shallow network to extract spectral features first. It serves as a basic spectral–spatial vision transformer for comparison.
7) *SSFTT [57]:* SSFTT is an improved transformer that uses a Gaussian-weighted feature tokenizer to generate input tokens for transformer blocks.
8) *T-CNN:* T-CNN utilizes a transfer learning strategy and attention mechanism to train a vanilla CNN, derived from [82], and is used to demonstrate the effectiveness of proposed transformer-based transfer learning methods.
9) *CT-Mixer [18]:* It is an improved transformer that effectively incorporates CNN into the transformer block's design, representing state-of-the-art transformer-based methods.

The comparison methods encompass a broad range of methods, including traditional methods, CNN-based techniques, transformer-based models, attention-based methods, and transfer learning methods. These selected methods can provide a comprehensive and thorough comparison. One can refer to the related papers for detailed experimental settings.

We use the iBOT pretrained ViT-B/16 as the foundation model to be fine-tuned, whose weights can be downloaded from https://github.com/bytedance/ibot. The iBOT utilizes contrastive self-distillation and MIM strategy to pretrain the given transformer which can be adapted to the downstream tasks. The main settings for ViT-B/16 are described as follows, input image size: 224 × 224 × 3, patch size: 16 × 16, embedding dimension: 768, number

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of blocks: 12, number of heads: 12, hidden dimension in MLP: 3072.

When using SS-VFMT and SS-VFMT-D, the input HSI images are generated with a spatial size of 27×27 to utilize spatial information. The spatial partition size is set to three, that is $P' = 3$. The hidden dimensions for the SpaEM and SpeEM are both set to 128. During training, a multistep learning rate scheduler is employed. The initial learning rate is 0.0004, and it is divided by ten after 80 and 160 epochs. The total number of epochs is set to 180 for all the three datasets. The mini-batch strategy with a batch size of 128 is used for all the designed methods.

In the experiments, the classification performance is primarily assessed based on overall accuracy (OA), average accuracy (AA), and Kappa coefficient (K). To ensure the robustness of the results, the experiments are conducted ten times with varying random initializations. In addition, the training samples are randomly selected from all labeled samples for each run.

C. Classification Results

The quantitative classification results for all three datasets using different methods are shown in Tables IV–VI. From these results, it can be seen that the proposed SS-VFMT and SS-VFMT-D have gained superior performance over other comparison methods in the case of twenty training samples per class. Specifically, these results show that

1) The CNN-based methods generally achieve better classification results than the transformer-based methods. As an example, on the Pavia University dataset, the CNN performs better than the SS-Tr with an improvement of 10.69% points for OA, 12.76% points for AA, and 0.1365 for K. Besides, in Table V, FDSSC yields better results than the CT-Mixer, with an improvement of 1.74% points for OA, 0.55% points for AA, and 0.0224 for K. There appears to be significant potential for improving transformer-based methods, particularly when working with a limited number of training samples.
2) Utilizing the well pretrained model can benefit the classification performance. Compared to CNN, T-CNN improves the classification results by 0.70% points, 0.52% points, and 0.0076 in terms of OA, AA, and K on the Houston dataset.
3) The proposed SS-VFMT obtains better classification performance than the comparison methods. Taking the Pavia University dataset as an example, SS-FT achieves better results than FDSSC, T-CNN, and CT-Mixer, with improvements of 0.66% points, 0.62% points, and 2.4% points in terms of OA, respectively.
4) The proposed SS-VFMT-D achieves the highest accuracies among all the comparison methods on the three datasets, which shows the effectiveness of the proposed architectures. The OAs obtained by the SS-VFMT-D on the three datasets outperform the ones of SS-VFMT by 0.91% points, 0.89% points, and 0.52% points, respectively. The distillation loss helps SS-VFMT obtain better classification performance.

D. Ablation Experiments

As we can see, PCA-based token generation (PTG), SpaEM, and SpeEM are three key components that make up the proposed methods. Therefore, the related ablation experiments are conducted to demonstrate the contribution and effectiveness of different modules. Generally, there are six models with different modules to be evaluated in this section as follows.

1) *Model-1:* **PTG** + ViT-B/16 + Linear. Only the linear layer is optimized during training.
2) *Model-2* **SpaEM-FM** + ViT-B/16 + Linear. The SpaEM and the linear layer are optimized during training.
3) *Model-3:* **PTG** + **SpaEM** + ViT-B/16 + Linear. The SpaEM and the linear layer are optimized during training.
4) *Model-4:* **PTG** + **SpeEM** + ViT-B/16 + Linear. The SpeEM and the linear layer are optimized during training.
5) *Model-5:* **SpaEM** + **SpeEM** + ViT-B/16 + Linear. The SpaEM, the SpeEM, and the linear layer are optimized during training.
6) *Proposed SS-VFMT:* **PTG** + **SpaEM** + **SpeEM** + ViT-B/16+ Linear.

The detailed results of the three datasets are shown in Table VII. From these results, it can be seen that different modules lead to different degrees of performance gain and each module can contribute to better classification results. Specifically, Model-1 and Model-2 are two naïve ideas which use PCA or shallow CNN to adapt a vision transformer-based foundation model to HSI classification. However, Model-1 obtains the worst classification performance in terms of OA, AA, and K on the three datasets. When compared with Model-1, Model-2 obtained higher accuracies, yet still much lower than the accuracies of the comparison methods. It shows that the simple adaption of foundation models (e.g., iBOT-pretrained vision transformer) is not enough to obtain a well-behaved HSI classification model. When comparing the classification accuracies of Model-3 and SS-VFMT, it can be seen that the SpeEM is very helpful for improving classification performance. For example, the OAs obtained by SS-VFMT outperform the ones of Model-2 by 4.67% points, 5.00% points, and 4.32% points on the Indian Pines, Pavia University, and Houston datasets, respectively. When comparing the classification results of Model-4 and SS-VFMT, it is obvious that the SpaEM leads to improvements of 6.82% points, 7.68% points, and 9.96% points in terms of OA on the three datasets, respectively. These experimental results verify the contribution of the SpaEM. Finally, Model-5 obtains lower classification accuracies than SS-VFMT, which demonstrates that it is useful to use PCA and generate the matched inputs for the patch embedding layer in the foundation transformer.

The ablation study using PCA or RGB to generate tokens is conducted to demonstrate the advantage of using PCA. The classification results on the three datasets have been shown in Table VIII. From Table VIII, it can be seen that using an RGB selection from the HSI degrades the classification performance of the proposed SS-VFMT when compared with using PCA to generate tokens. Specifically, the OAs obtained by using

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TABLE IV
TESTING DATA CLASSIFICATION RESULTS (VALUES ± STANDARD DEVIATION) ON THE INDIAN PINES DATASET

Class	EMP-SVM	CNN	SSRN	DBDA	FDSSC	SS-Tr	SSFTT	T-CNN	CT-Mixer	**SS-VFMT**	**SS-VFMT-D**
	96.54±2.07	100.0±0.00	86.04±9.73	79.35±12.80	89.04±10.60	99.62±1.15	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00
	63.74±7.45	78.96±6.22	86.85±6.00	86.89±10.74	92.35±6.46	68.00±5.20	85.38±3.92	85.20±3.83	81.94±3.95	82.14±4.39	83.00±4.29
	76.56±4.40	87.75±5.96	86.54±5.65	86.88±8.79	87.91±11.64	82.05±4.23	87.63±5.69	87.37±2.82	87.47±4.29	91.12±5.20	90.12±5.03
	81.94±5.14	98.53±2.56	73.29±13.10	81.64±11.47	80.76±14.64	96.96±3.15	98.06±2.43	97.83±2.95	99.35±1.11	98.94±2.36	99.63±0.68
	86.57±3.85	92.18±3.21	98.00±1.96	97.02±3.74	98.69±1.36	87.21±4.52	94.15±2.09	91.88±2.37	93.24±1.54	93.82±1.73	94.06±1.71
	92.85±4.63	95.41±2.53	97.79±1.40	96.54±1.25	97.95±1.56	89.54±6.27	97.98±1.26	93.82±3.31	95.30±3.47	96.83±4.60	96.96±3.07
	92.50±6.12	100.0±0.00	73.78±18.70	58.08±28.22	66.78±30.24	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00
	94.10±3.25	99.80±0.59	99.46±0.83	99.62±0.58	99.93±0.10	99.85±0.31	99.34±1.22	99.76±0.46	99.98±0.07	100.0±0.00	100.0±0.00
	98.00±6.00	100.0±0.00	44.13±20.56	32.65±11.22	32.03±21.44	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00
	68.28±7.20	88.93±3.10	79.53±6.45	81.08±8.08	79.27±12.02	80.02±6.00	89.66±4.47	88.76±4.87	87.88±4.44	89.87±3.46	90.68±4.00
	59.22±4.69	87.41±5.47	92.94±3.50	95.43±4.19	92.20±6.64	75.59±5.37	84.72±5.24	86.29±3.66	86.89±3.17	88.12±5.85	90.60±4.39
	66.61±7.68	83.84±6.23	82.78±10.73	90.57±13.45	87.94±14.32	74.31±4.72	77.40±4.99	83.00±5.36	82.57±6.87	93.18±2.86	93.56±3.94
	97.08±1.81	99.57±0.89	95.49±3.82	91.42±5.35	92.61±4.98	94.11±4.43	98.86±2.15	99.19±1.57	98.97±1.89	99.95±0.16	99.78±0.36
	86.39±5.68	97.39±1.71	97.87±1.14	98.43±1.05	98.57±8.41	93.16±3.39	98.75±1.18	98.34±1.52	97.07±1.48	98.09±2.20	99.19±0.80
	71.48±8.55	97.19±2.60	87.30±7.75	88.99±5.46	83.25±12.18	92.35±6.11	96.91±2.02	97.70±2.93	96.97±2.59	99.43±1.46	99.40±1.71
	95.48±4.71	99.18±0.91	79.51±4.48	68.09±12.75	80.42±6.41	94.93±4.02	100.0±0.00	99.45±0.67	99.59±0.63	99.45±1.10	100.0±0.00
OA (%)	73.32±2.25	89.62±1.72	89.44±1.38	90.26±3.06	89.71±2.72	81.92±2.47	90.10±1.58	90.26±1.69	89.87±1.28	91.67±1.97	**92.58±1.45**
AA (%)	82.96±1.31	94.14±0.72	85.08±2.08	83.29±2.47	84.98±3.97	89.23±1.46	94.30±0.83	94.29±0.89	94.20±0.72	95.68±0.88	**96.06±0.79**
K×100	69.93±2.50	88.20±1.93	88.00±1.55	88.95±3.43	88.30±3.08	79.59±2.74	88.76±1.78	88.94±1.90	88.49±1.44	90.52±2.21	**91.55±1.63**

TABLE V
TESTING DATA CLASSIFICATION RESULTS (VALUES ± STANDARD DEVIATION) ON THE PAVIA UNIVERSITY DATASET

Class	EMP-SVM	CNN	SSRN	DBDA	FDSSC	SS-Tr	SSFTT	T-CNN	CT-Mixer	**SS-VFMT**	**SS-VFMT-D**
	81.27±6.60	90.85±4.53	97.84±1.71	98.74±1.20	98.59±1.61	76.43±7.00	86.16±6.15	93.33±4.23	90.59±5.26	94.82±5.02	95.92±3.66
	83.13±3.26	93.83±3.49	97.72±0.81	99.51±0.36	99.19±0.38	84.77±4.48	94.50±3.60	95.77±2.98	94.62±4.88	95.47±4.34	96.97±4.03
	81.60±4.51	98.12±1.27	83.71±8.17	90.81±12.10	92.84±5.72	88.63±3.30	93.94±4.13	98.48±0.77	94.39±4.19	98.66±1.57	98.68±2.10
	95.29±2.44	96.29±1.39	97.70±2.01	92.74±7.92	94.75±5.75	67.14±7.81	88.97±5.25	92.77±1.87	84.61±6.80	97.72±0.97	96.92±1.23
	99.26±0.26	99.33±0.52	99.86±0.27	99.53±0.63	99.88±0.12	89.37±7.60	98.95±1.06	99.15±0.60	99.18±0.67	100.0±0.00	100.0±0.00
	80.27±6.31	99.47±0.63	91.98±3.69	90.96±5.45	95.65±2.21	97.29±2.48	96.07±3.90	99.45±0.97	99.53±1.08	99.47±0.44	99.73±0.35
	93.11±1.56	99.39±0.68	88.49±12.03	93.80±8.83	96.36±2.67	95.35±3.62	99.58±0.67	99.37±0.66	99.34±0.67	99.85±0.46	99.87±0.39
	83.86±3.96	98.94±0.80	84.79±7.36	89.83±6.47	83.63±11.84	91.74±7.70	86.63±7.99	98.52±1.04	97.10±2.75	97.87±1.46	98.83±0.79
	99.85±0.12	96.66±1.33	99.41±0.94	96.82±1.69	97.35±3.44	67.41±7.45	95.54±2.17	91.77±2.47	93.95±3.14	99.82±0.44	99.92±0.05
OA (%)	84.53±2.22	95.26±1.74	94.72±1.17	95.87±1.85	96.07±1.89	84.57±2.78	92.61±1.92	96.11±1.49	94.33±2.85	96.73±2.39	**97.62±2.14**
AA (%)	88.63±1.57	96.99±0.74	93.50±2.07	94.75±2.36	95.36±1.64	84.23±1.62	93.37±1.24	96.51±0.63	94.81±1.77	98.19±0.90	**98.54±0.82**
K×100	80.00±2.76	93.80±2.23	93.02±1.53	94.58±2.41	94.84±2.42	80.15±3.39	90.29±2.47	94.89±1.93	92.60±3.63	95.73±3.07	**96.88±2.76**

PCA on the three datasets outperform the ones by using RGB selection by 2.07% points, 1.21% points, and 0.62% points, respectively.

We also conduct an ablation study about using PCA-processed or original HSI samples to generate spatial enhancement tokens. The classification results on the three datasets have been shown in Table IX. From Table IX, it can be seen that using PCA-processed samples to generate spatial enhancement tokens degrades the classification performance of the proposed SS-VFMT. Specifically, the OAs obtained using original HSI input on the three datasets outperform the ones by using PCA-processed input by 2.9% points, 6.0% points, and 1.96% points, respectively. We infer the reasons include that i) the use of the original HSI image leads to more informative inputs compared to the PCA which retains the three principal components. ii) If the SpaEM uses PCA to form input, the two parts of the transformer input tokens will tend to be homogeneous, in which case the SpaEM will lose its significance, and the use of spatial enhancement tokens to reweight the pretrained attention score will be much less effective.

To help understand the $\mathbf{A}_Q$, $\mathbf{A}_K$, and $\mathbf{A}_V$ matrices, we conduct an ablation study to verify the advantage of using the center patch's spectral feature to obtain $\mathbf{A}_Q$ and $\mathbf{A}_K$, rather than directly using $\mathbf{F}$. Table X shows the classification results. From the results, one can see that setting all the three matrices $\mathbf{A}_Q$, $\mathbf{A}_K$, and $\mathbf{A}_V$ to $\mathbf{F}$ significantly degrades the classification performance of the proposed SS-VFMT, when compared with the original generation process. Specifically, setting all three matrices to F decreases the OAs of the proposed SS-VFMT on the three datasets by 5.84% points, 6.78% points, and 6.93% points, respectively.

Then, we turn our attention to the patch-level relationship distillation. As for the SS-VFMT-D, the distillation loss will influence the classification performance. Therefore, in the following part, the ablation experiments about the distillation loss are conducted to demonstrate its contribution.

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TABLE VI

TESTING DATA CLASSIFICATION RESULTS (VALUES ± STANDARD DEVIATION) ON THE HOUSTON DATASET

Class	EMP-SVM	CNN	SSRN	DBDA	FDSSC	SS-Tr	SSFTT	T-CNN	CT-Mixer	**SS-VFMT**	**SS-VFMT-D**
	92.99±4.30	92.59±4.56	96.25±2.94	93.48±5.59	96.41±2.30	79.95±6.28	95.36±3.62	92.49±4.18	90.90±5.13	91.17±4.52	90.15±4.49
	93.06±5.72	97.00±2.21	97.65±2.48	95.10±3.78	97.64±2.11	75.20±7.35	98.58±1.13	95.74±1.98	95.11±2.60	93.14±5.08	95.81±1.96
	98.97±1.10	98.66±1.41	99.93±0.22	100.0±0.00	100.0±0.00	97.15±1.77	99.20±0.60	98.21±1.95	97.10±4.20	99.92±0.14	99.99±0.04
	94.75±2.94	97.66±1.81	95.98±4.13	97.13±2.17	95.63±3.79	60.92±8.77	97.82±3.19	95.71±2.10	92.27±4.20	96.96±2.56	97.37±2.36
	96.51±4.52	97.47±5.11	95.41±2.36	97.66±2.42	97.60±2.48	97.90±2.04	99.99±0.02	97.77±2.11	98.58±3.08	98.13±5.16	98.85±2.43
	94.72±3.42	95.38±3.52	97.31±7.83	97.37±2.23	99.80±0.34	91.97±4.82	98.42±3.86	94.62±3.53	96.42±3.53	95.50±4.57	94.52±3.83
	85.54±4.67	90.54±2.41	92.10±2.47	91.93±3.62	92.56±4.64	74.87±6.62	84.32±5.32	92.72±2.53	89.18±3.91	91.69±1.72	92.63±2.13
	69.36±4.90	78.48±6.64	93.23±3.49	94.88±3.19	92.68±3.66	78.24±4.24	80.80±5.19	80.82±3.17	83.90±4.83	84.98±4.07	83.79±3.74
	75.81±6.81	90.60±3.94	89.87±3.83	88.57±2.27	90.82±3.19	79.04±4.05	78.82±6.79	90.48±3.48	86.58±2.31	91.24±2.87	91.43±2.23
	87.63±4.01	96.06±4.12	86.49±6.78	89.76±3.83	89.12±4.70	96.12±2.61	95.10±3.33	99.20±0.90	99.09±1.52	98.56±1.77	99.49±0.60
	85.58±7.72	91.52±4.71	90.45±1.94	95.44±2.10	92.70±3.56	90.58±7.82	95.82±3.23	93.94±4.66	92.34±6.17	92.62±6.30	94.21±5.35
	76.18±6.14	91.81±5.48	89.91±4.73	93.15±3.27	93.40±3.86	84.73±5.04	89.65±4.87	93.76±5.57	91.27±5.74	89.83±6.33	90.24±5.72
	56.44±5.90	96.08±2.62	93.52±5.65	82.75±6.06	83.22±9.14	92.98±5.95	97.75±2.10	96.30±2.57	91.22±5.39	91.60±3.01	93.63±3.39
	97.94±2.53	99.93±0.22	97.67±3.29	98.13±2.59	98.07±2.72	100.0±0.00	99.85±0.22	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00
	99.08±0.46	99.59±1.06	96.93±1.96	95.05±2.47	96.22±3.18	96.84±2.96	100.0±0.00	99.94±0.67	98.63±2.38	99.98±0.05	99.98±0.05
OA (%)	86.56±1.36	93.36±0.92	93.32±1.05	93.67±0.92	93.97±0.99	84.13±1.72	92.88±0.97	94.06±0.97	92.73±1.21	93.68±1.24	**94.20±0.59**
AA (%)	86.97±1.26	94.23±0.67	94.18±1.12	94.03±0.87	94.39±1.00	86.43±1.54	94.10±0.85	94.75±0.70	93.51±1.04	94.36±1.05	**94.81±0.59**
K×100	85.47±1.47	92.82±0.99	92.78±1.13	93.16±1.00	93.48±1.07	82.86±1.86	92.30±1.05	93.58±1.04	92.14±1.31	93.16±1.34	**93.72±0.64**

TABLE VII

ABLATION EXPERIMENTS FOR DIFFERENT MODULES

Name	PTG	SpaEM	SpeEM	Metrics	Indian Pines	Pavia University	Houston
Model-1	✓	×	×	OA (%)	81.60±1.69	83.36±1.70	75.21±1.99
				AA (%)	89.36±1.05	83.57±1.22	83.57±1.73
				K×100	79.22±1.86	78.56±2.06	78.57±2.14
Model-2	×	✓	×	OA (%)	86.34±1.50	90.78±3.04	88.34±1.39
				AA (%)	92.07±0.88	89.06±1.77	89.96±1.08
				K×100	84.53±1.67	87.98±3.82	87.40±1.50
Model-3	✓	✓	×	OA (%)	87.00±1.40	91.73±2.70	89.36±1.41
				AA (%)	92.41±0.70	90.10±1.57	90.71±1.12
				K×100	85.28±1.55	89.18±3.42	88.50±1.52
Model-4	✓	×	✓	OA (%)	84.82±1.50	89.05±2.99	83.72±0.89
				AA (%)	91.27±0.72	86.82±2.12	85.76±0.78
				K×100	82.82±1.65	85.68±3.72	82.41±2.14
Model-5	×	✓	✓	OA (%)	89.26±1.85	95.57±2.89	89.72±1.86
				AA (%)	94.19±0.92	96.86±1.37	91.15±1.55
				K×100	87.81±2.06	94.22±3.71	88.89±2.01
SS-FMT	✓	✓	✓	OA (%)	91.67±1.97	96.73±2.39	93.68±1.24
				AA (%)	95.68±0.88	98.19±0.90	94.36±1.04
				K×100	90.52±2.21	95.73±3.07	93.16±1.34

TABLE VIII

CLASSIFICATION RESULTS OBTAINED USING PCA-BASED AND RGB-BASED TOKEN GENERATION FOR SS-VFMT

		Indian Pines	Pavia University	Houston
PCA	OA (%)	91.67±1.97	96.73±2.39	93.68±1.24
	AA (%)	95.68±0.88	98.19±0.90	94.36±1.05
	K×100	90.52±2.21	95.73±3.07	93.16±1.34
RGB	OA (%)	89.60±2.48	95.52±1.99	93.06±0.93
	AA (%)	94.54±1.13	97.21±1.10	93.90±0.88
	K×100	88.19±2.77	94.13±2.57	92.49±1.01

TABLE IX

CLASSIFICATION RESULTS OBTAINED USING PCA-PROCESSED AND ORIGINAL HSI SAMPLES FOR SPAEM OF SS-VFMT

		Indian Pines	Pavia University	Houston
Original	OA (%)	91.67±1.97	96.73±2.39	93.68±1.24
	AA (%)	95.68±0.88	98.19±0.90	94.36±1.05
	K×100	90.52±2.21	95.73±3.07	93.16±1.34
PCA	OA (%)	88.77±1.93	90.73±1.04	91.72±1.60
	AA (%)	94.19±1.01	95.73±0.86	92.76±1.40
	K×100	87.24±2.16	88.05±1.27	91.05±1.73

Fig. 10 shows the OAs obtained by SS-VFMT-D with different values of loss ratio α. It can be seen that a proper value of α is needed. Generally speaking, a larger value of α limits the model by the spatial relationship obtained using PCA tokens, which is fixed during training. That is, the model will pay more attention to the spatial information in HIS, which will harm the whole model's spectral–spatial feature extraction ability. On the contrary, a smaller value of α forces the model to degrade into SS-VFMT, where the enhancement modules cannot learn the spatial relationship from the pretrained knowledge well. Specifically, for the Indian Pines dataset, the best OA is obtained when the value is set to 0.02. The optimal

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TABLE X
CLASSIFICATION RESULTS OF SS-VFMT OBTAINED USING DIFFERENT MATRICES GENERATION METHODS

		Indian Pines	Pavia University	Houston
Original	OA (%)	91.67±1.97	96.73±2.39	93.68±1.24
	AA (%)	95.68±0.88	98.19±0.90	94.36±1.05
	K×100	90.52±2.21	95.73±3.07	93.16±1.34
All **A** matrices are **F**	OA (%)	85.83±2.05	89.95±2.67	86.75±1.71
	AA (%)	91.72±1.23	88.76±1.52	88.55±1.53
	K×100	83.96±2.27	86.90±3.33	85.67±1.85

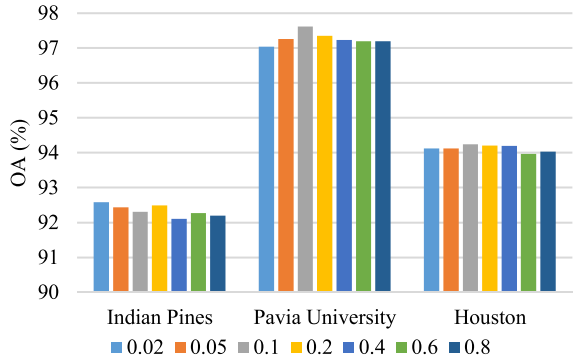


Fig. 10. OA of SS-VFMT-D with different values of loss ratio α.

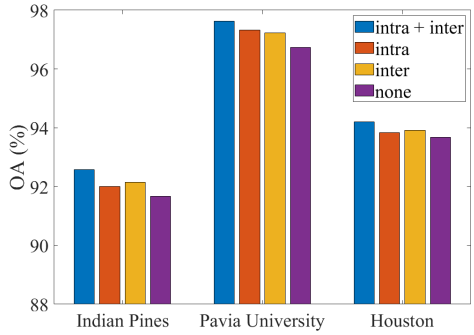


Fig. 11. OA of SS-VFMT-D with respect to different kinds of distillation losses.

value of α for the Pavia University dataset is 0.1, the same as the Houston dataset.

Fig. 11 illustrates the overall accuracies obtained by SS-VFMT-D using different types of distillation loss. It can be seen that either intraimage distillation loss or interimage distillation has a different degree of positive contribution to the SS-VFMT-D. The combination of intraimage patch-level distillation loss and interimage patch-level distillation loss achieves the best classification performance.

E. Algorithm Complexity

Table XI shows the number of FLOPs, parameters, training, and testing time for different methods on the Pavia University Dataset. As Model-1 and Model-2 are both typical ways to adapt a pretrained model to an HSI classification task, it is also used for comparison. "$*$" means that the weights in the pretrained transformer are not frozen during training. FLOPs, which means the number of floating-point operations needed for a given model, are used here to measure the amount of calculation in the feedforward propagation. Param. means the number of parameters to be trained in a given model. Therefore, the number of learning parameters is also calculated. Besides, the training and testing time are used here to measure the complexity of different algorithms.

From the results, it can be seen that the adaption of a transformer-based foundation model usually results in a heavier computational burden than other comparison methods. The reason is that the transformers used in foundation models for general CV tasks are usually bigger than those designed in HSI tasks. The number of FLOPs and parameters of SS-VFMT are both the highest among these methods, while its overall classification accuracy is the highest. Besides, the number of learnable parameters is much smaller than that of the SS-VFMT, due to the strategy of freezing transformers. Thus, the training time and testing time of SS-VFMT are smaller than those of SS-VFMT$*$. When comparing SS-VFMT with Model-1 and Model-2, it can be seen that their numbers of total parameters are nearly the same, due to the fact that the spatial and SpeEM is lightweight. The number of FLOPs of SS-VFMT is almost equal to the addition of Model-1 and Model-2, due to the fact that the computation is mainly caused by the processing of PCA tokens and the spatial enhancement tokens.

F. Classification Maps

Figs. 12–14 show the classification maps for different methods over the three datasets. From these qualitative results, it can be clearly seen that the proposed SS-VFMT and SS-VFMT-D have achieved more competitive classification performance than other methods. For example, on the Pavia University dataset, many methods tend to confuse some *Metal Sheets* and *Shadow* regions because they are located close to each other. Many methods tend to predict the pixels in *Shadow* regions as *Metal Sheets*. We infer that these models are overfitting to this type of spatial information while neglecting the spectral information. However, the proposed methods perform well in this area, meaning that they can utilize spectral and spatial well for better classification results.

VI. EXPERIMENTS OF HSI ZERO-SHOT CLASSIFICATION

This section of the experiments is mainly about using the proposed SS-VLFMT for GZSL in HSI classification.

A. Datasets Description of SS-VLFMT

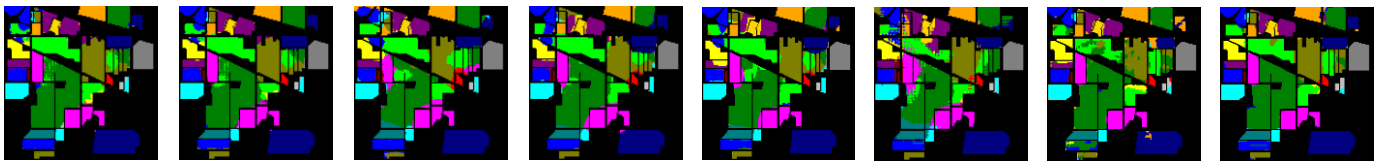
For the GZSL in the HSI classification, the proposed SS-VLFMT is evaluated on the three HSI datasets. In order to reduce the difficulty of zero-sample classification, we merge some subclasses in the Indian Pines and Houston datasets. Specifically, the merged Indian Pines dataset contains ten different land-cover types, Fig. 15 shows the false color map and corresponding ground-truth map of the merged Indian Pines dataset. Furthermore, Table XII presents the number of training and test samples per class. The merged Houston dataset labeled 12 different land-cover types. Fig. 16 depicts the false color map and corresponding ground-truth map, while Table XIII provides the number of training and test samples per

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TABLE XI
FLOPs, Parameters, Training, and Testing Time for Different Methods on the Pavia University Dataset

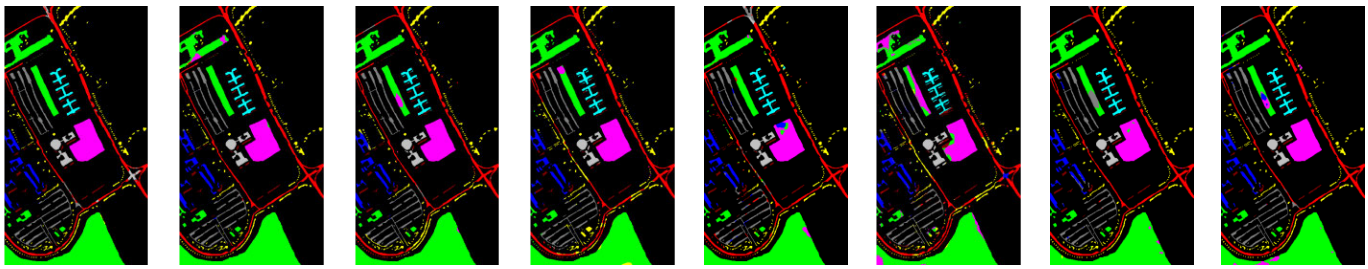
	FDSSC	T-CNN	SS-Tr	SSFTT	CT-Mixer	Model-1*	Model-1	Model-2*	Model-2	SS-VFMT*	SS-VFMT
FLOPs	137.11M	37.01M	84.22M	11.71M	905.44M	17.58G	17.58G	7.29G	7.29G	25.38G	25.38G
Total Param.	0.65M	0.27M	0.7M	0.2M	0.8M	85.9M	85.9M	86.9M	86.9M	87.0M	87.0M
Learnable Param.	0.65M	0.27M	0.7M	0.2M	0.8M	85.9M	0.01M	87.0M	1.06M	87.0M	1.17M
Training time (s)	30.42	7.65	11.28	1.93	43.88	256.42	95.88	109.73	76.75	427.15	311.21
Testing time (s)	11.85	9.32	10.27	0.99	27.05	108.37	108.77	52.31	50.33	179.25	178.42
OA (%)	96.07	96.11	84.57	92.61	94.33	82.25	83.36	81.76	90.78	88.19	96.73

"*" means that the weights in the pre-trained Transformer are also optimized in the training.



(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 12. Classification maps using different methods on the Indian Pines dataset. (a) SS-VFMT-D, (b) SS-VFMT, (c) T-CNN, (d) CT-Mixer, (e) SSFTT, (f) SS-Tr, (g) FDSCC, and (h) DBDA.



(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 13. Classification maps using different methods on the Pavia University dataset. (a) SS-VFMT-D, (b) SS-VFMT, (c) T-CNN, (d) CT-Mixer, (e) SSFTT, (f) SS-Tr, (g) FDSCC, and (h) DBDA.

TABLE XII
Land Cover Classes and Numbers of Samples for the Merged Indian Pines Dataset

No.	Class Name	Training samples	Test samples	Total samples
1	Alfalfa	20	26	46
2	Corn	20	2475	2495
3	Grass	20	1221	1241
4	Hay-windrowed	20	458	478
5	Oats	15	5	20
6	Soybean	20	4000	4020
7	Wheat	20	185	205
8	Woods	20	1245	1265
9	Buildings-Grass-Trees	20	366	386
10	Stone-Steel-Towers	20	73	93
	Total	195	10054	10249

class. The details and settings of the Pavia University dataset are consistent with those in Section V-A.

B. Experimental Setup of SS-VLFMT

The CLIP's text encoder is used in the text branch for SS-VLFMT, and the corresponding output dimension is 512. For the hash code length, we take $L = 8$. The weighted coefficients β and γ are both set to be 3.

For the ZSL setting, two classes from the whole classes are selected as unseen classes on the three datasets. The training settings are set to be almost the same as SS-VFMT.

As for the evaluation metric, the model needs to recognize HSI pixels from both seen and unseen classes in the testing stage. Following [83], we compute the AAs of both seen classes (S) and unseen classes (U). Because high performance in densely populated classes is encouraged when using OA. Besides, the harmonic mean accuracy (H), which takes both the seen and unseen classes' accuracies into consideration, is introduced here

$$\mathrm{H} = \frac{2}{\frac{1}{\mathrm{S}} + \frac{1}{\mathrm{U}}} = \frac{2 \times \mathrm{S} \times \mathrm{U}}{\mathrm{S} + \mathrm{U}}. \tag{55}$$

C. Classification Results of SS-VLFMT

There are too many combinations when selecting two classes as unseen classes form all the classes in the training. Therefore, three specific combinations are selected from all unknown classes' combinations to demonstrate the classification results for each dataset.

The classification results are presented in Tables XIV–XVI. Each of these tables' first rows gives which two classes are unseen during training. From the results, one can see that 1) it is evident that the traditional HSI classification methods exhibit a limitation in identifying unseen classes, as indicated

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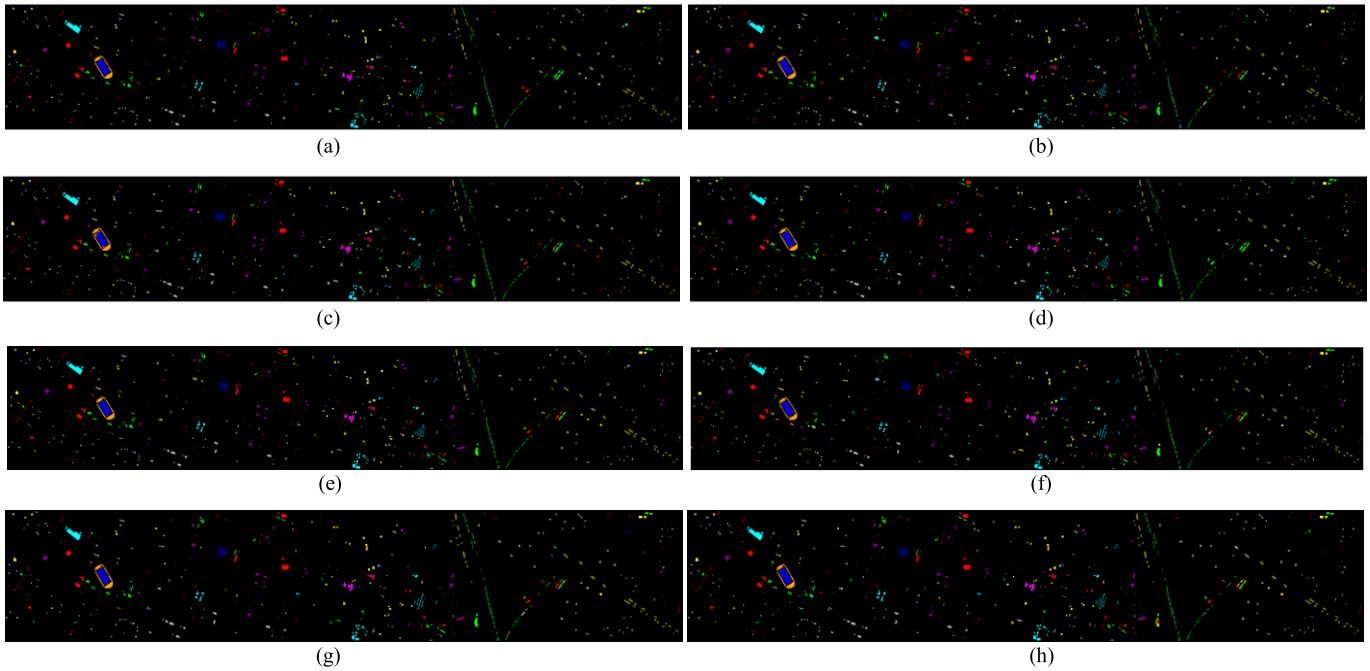


Fig. 14. Classification maps using different methods on the Houston dataset. (a) SS-VFMT-D, (b) SS-VFMT, (c) T-CNN, (d) CT-Mixer, (e) SSFTT, (f) SS-Tr, (g) FDSCC, and (h) DBDA.

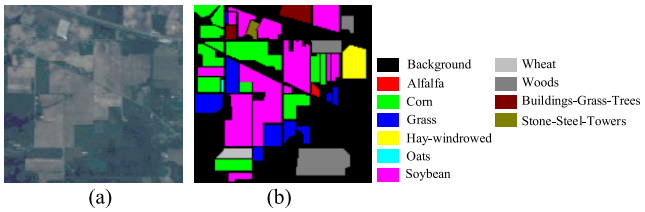


Fig. 15. Merged Indian Pines dataset: (a) false color map and (b) ground-truth.

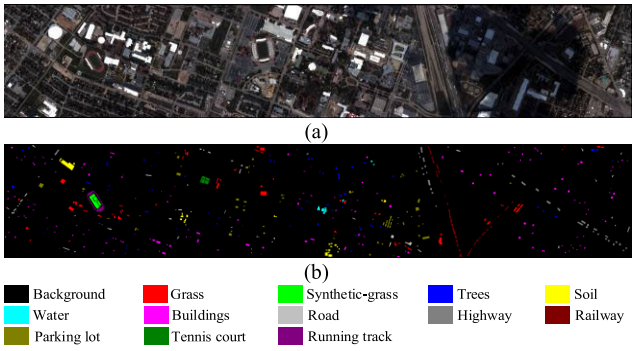


Fig. 16. Merged Houston dataset: (a) false color map and (b) ground-truth.

by their zero accuracies for these classes. However, the proposed SS-VLFMT has the potential to recognize samples from unseen classes due to the specially designed scheme for ZSL. 2) Recognition of samples from unseen classes may slightly reduce their effectiveness in classifying samples from seen classes. The average accuracies of seen classes obtained by the proposed SS-VLFMT are lower than that of traditional methods. For example, in Table XV, when the eighth and ninth classes (i.e., bricks and shadow) are unseen in the training on the Pavia University dataset, and the accuracy of seen classes (S) obtained by SS-VLFMT is 93.79%, which is lower than 97.30% obtained by CT-Mixer and 97.86% obtained by DBDA. The class accuracies of most seen classes obtained using SS-VLFMT are lower than those of CT-Mixer and DBDA. 3) Different combinations of unseen classes often lead to different generalized zero-shot classification performances. For example, on the Indian Pines dataset, when the second and fouth classes (i.e., corn and hay-windrowed) are set as unseen classes during training, their class accuracies are 23.49% and 92.26%, respectively. When the fourth and tenth classes (i.e., hay-windrowed and stone-steel-towers) are set as unseen classes during training, the class accuracy of the fourth class is higher, but the samples of the other class are almost unrecognizable, with a class accuracy of zero. For the latter unseen class combinations in Table XIV, the class accuracies of the two unseen classes are both low.

D. Discussion of Unseen Classes Selection

In this section, we first analyze the conditions under which zero-shot classification may succeed, beginning with an overview of the proposed zero-shot classification mechanism. Based on this analysis, we provide guidelines for selecting specific unseen classes and then conduct relevant experiments.

As illustrated in Fig. 17(a), the model lacks access to samples from unseen classes during training, which prevents it from directly establishing a connection between the semantic names of unseen classes and their corresponding image samples [represented by the green arrow in Fig. 17(a)]. Instead, the model infers the relationship between the image features of unseen classes and their semantic features (derived from class names) through three established relationships: 1) the relationship between the image features and semantic features

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TABLE XIII
LAND COVER CLASSES AND NUMBERS OF SAMPLES FOR THE MERGED HOUSTON DATASET

No.	Class Name	Training samples	Test samples	Total samples
1	Grass	20	2485	2505
2	Synthetic grass	20	677	697
3	Trees	20	1224	1244
4	Soil	20	1222	1242
5	Water	20	305	325
6	Buildings	20	2492	2512
7	Road	20	1232	1252
8	Highway	20	1207	1227
9	Railway	20	1215	1235
10	Parking lot	20	1682	1702
11	Tennis court	20	408	428
12	Running track	20	640	660
	Total	240	14789	15029

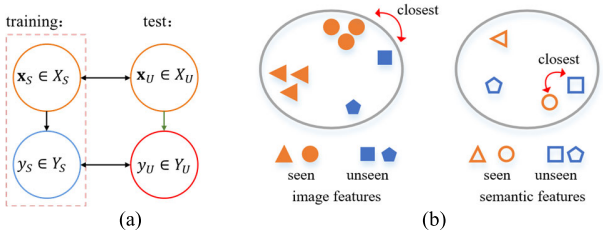


Fig. 17. Illustration of zero-shot classification: (a) training and test process and (b) image and semantic feature space.

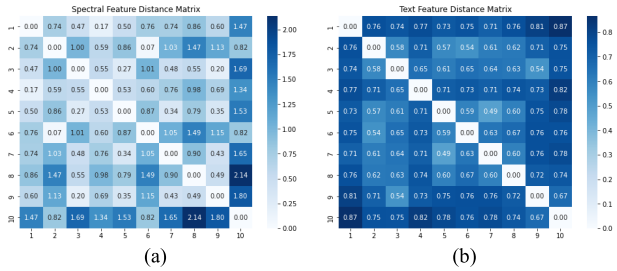


Fig. 18. Interclass distances of the merged Indian Pines dataset: (a) spectral distance and (b) semantic distance.

of seen classes; 2) the relationship between the image features of seen and unseen classes; and 3) the relationship between the semantic features of seen and unseen classes. The first relationship can be learned through image–text alignment during model training, while the effectiveness of ZSL depends on the latter two relationships. If the neighboring classes of any given unseen class in the image feature space remain neighbors in the semantic space, as shown in Fig. 17(b), the model is more likely to perform well.

From this analysis, we can select classes that meet the following criteria as unseen classes for experiments: 1) the image features of the unseen classes should not closely resemble those of other classes and 2) the neighboring seen classes of the selected unseen classes in the image feature space should retain their corresponding relationships in the semantic space. The first criterion facilitates the model's ability to discover new classes, while the second allows the model to accurately identify the names of these newly discovered classes.

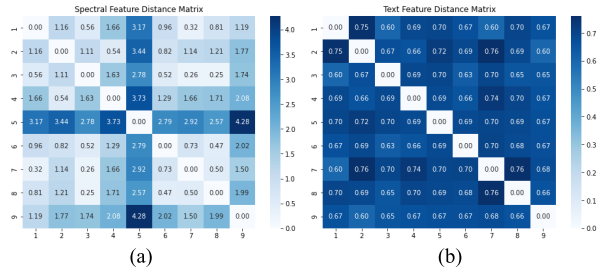


Fig. 19. Interclass distances of the Pavia University dataset: (a) spectral distance and (b) semantic distance.

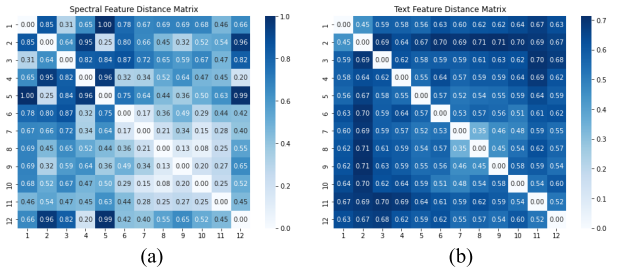


Fig. 20. Interclass distances of the merged Houston dataset: (a) spectral distance and (b) semantic distance.

For simplicity, we use the mean spectral features and semantic features to calculate the Euclidean distances between different classes.

Fig. 18 presents the interclass distances within the merged Indian Pines dataset. Considering the first and second classes as unseen classes, it is evident that the class most similar to the first class in the spectral feature space is the fourth class. However, in the semantic space, the text feature of the fourth class is more similar to that of the second class than to the first class, which may lead to an increase in incorrect predictions. The mean spectral feature of the second class is also too similar to that of the sixth class, making it challenging for the model to recognize samples from the second class accurately. Indeed, the classification results in Table XVII confirm this analysis, as both the first and second classes fail to achieve high accuracy. When we consider the first and ninth classes as unseen classes, it is apparent that the class most similar to the first class is the fourth class based on spectral feature distances depicted in Fig. 18(a), while the text feature of the fourth class is more similar to that of the ninth class in Fig. 18(b). This inconsistency suggests that the neighborhood relationships for the first class differ across the two feature spaces. For the ninth class, the third class is most similar in the spectral feature space, and in the semantic space, the text feature of the third class is more similar to that of the ninth class than to that of the first class. Thus, the neighborhood relationships in both image feature and semantic spaces are consistent for the ninth class. As indicated in Table XVII, while the accuracy for the first class is low, the accuracy for the ninth class is significantly higher. In addition, Table XVII records the experimental results of various combinations for further analysis.

Fig. 19 illustrates the interclass distances for the Pavia University dataset. When considering the first and fourth

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TABLE XIV

TESTING DATA CLASSIFICATION RESULTS ON THE MERGED INDIAN PINES DATASET

	(2,4)*			(4,10)			(6,8)		
	CT-Mixer	DBDA	SS-VLFMT	CT-Mixer	DBDA	SS-VLFMT	CT-Mixer	DBDA	SS-VLFMT
	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
	0.00	0.00	23.49	74.95	68.48	71.99	95.52	90.95	81.86
	91.24	93.04	91.89	94.10	96.40	87.31	96.31	95.09	89.93
	0.00	0.00	92.26	0.00	0.00	83.89	100.0	100.0	100.0
	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
	92.76	82.10	88.90	85.88	75.25	75.86	0.00	0.00	16.87
	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
	91.97	99.28	91.57	95.42	95.10	89.08	0.00	0.00	13.44
	93.17	100.0	100.0	98.63	100.	75.97	98.09	98.91	100.0
	100.0	100.0	100.0	0.00	0.00	0.00	100.0	100.0	100.0
OA (%)	65.37	62.52	**74.30**	**81.27**	75.77	77.89	46.03	44.76	**50.38**
AA (%)	76.91	77.44	**88.81**	74.90	73.53	**78.31**	78.99	78.41	**80.21**
K×100	52.80	50.74	**64.87**	**75.41**	68.51	70.90	34.34	33.48	**38.76**
S (%)	96.14	**96.80**	96.54	**93.62**	91.90	87.40	**98.74**	98.02	96.47
U (%)	0.00	0.00	**57.87**	0.00	0.00	**41.95**	0.00	0.00	**15.15**
H×100	0.00	0.00	**72.37**	0.00	0.00	**56.69**	0.00	0.00	**26.19**

* (2, 4) means that the second and fourth classes are unseen during training, and the same applies thereafter.

TABLE XV

TESTING DATA CLASSIFICATION RESULTS ON THE PAVIA UNIVERSITY DATASET

	(8,9)			(4,5)			(4,9)		
	CT-Mixer	DBDA	SS-VLFMT	CT-Mixer	DBDA	SS-VLFMT	CT-Mixer	DBDA	SS-VLFMT
	91.95	98.87	88.17	92.00	98.46	91.15	88.63	91.03	81.67
	96.05	93.11	90.86	96.07	96.41	87.97	95.96	96.50	91.51
	99.33	99.86	97.74	95.57	99.13	95.96	86.44	97.88	99.95
	94.22	93.20	97.40	0.00	0.00	23.24	0.00	0.00	8.09
	100.0	100.0	100.0	0.00	0.00	34.13	99.85	100.0	100.0
	99.54	99.98	96.43	98.94	100.0	95.37	99.72	99.84	98.66
	100.0	100.0	85.96	100.0	100.0	100.0	100.0	100.0	100.0
	0.00	0.00	40.90	99.51	99.75	95.49	99.10	99.73	97.92
	0.00	0.00	40.76	100.0	98.92	100.0	0.00	0.00	54.91
OA (%)	85.67	85.46	**86.61**	86.32	**87.76**	84.65	86.29	**87.53**	85.51
AA (%)	75.68	76.12	**82.02**	75.79	76.96	**80.37**	74.41	76.11	**81.41**
K×100	81.19	80.99	**82.67**	81.76	**83.62**	80.03	81.52	**83.23**	81.14
S (%)	97.30	**97.86**	93.79	97.44	**98.95**	95.13	95.66	**97.85**	95.67
U (%)	0.00	0.00	**40.83**	0.00	0.00	**28.68**	0.00	0.00	**31.50**
H×100	0.00	0.00	**56.89**	0.00	0.00	**44.08**	0.00	0.00	**47.40**

TABLE XVI

TESTING DATA CLASSIFICATION RESULTS ON THE MERGED HOUSTON DATASET

	(6,11)			(5,6)			(3,4)		
	CT-Mixer	DBDA	SS-VLFMT	CT-Mixer	DBDA	SS-VLFMT	CT-Mixer	DBDA	SS-VLFMT
	83.66	83.58	81.09	85.92	84.71	84.91	85.51	81.77	86.44
	99.11	99.26	100.0	99.11	99.11	99.85	98.82	99.56	99.56
	100.0	100.0	96.72	99.92	99.35	99.92	0.00	0.00	12.22
	99.75	100.0	100.0	100.0	100.0	99.75	0.00	0.00	27.78
	98.03	98.03	95.08	0.00	0.00	31.38	98.03	97.38	91.80
	0.00	0.00	16.56	0.00	0.00	33.08	74.48	73.233	68.06
	80.44	84.50	81.01	86.04	83.85	88.64	85.63	84.82	86.85
	92.71	79.70	93.70	100.0	84.26	93.45	93.37	82.44	87.57
	90.21	91.28	94.49	91.69	97.12	94.98	88.48	90.62	95.23
	91.38	87.40	69.03	94.29	89.48	68.79	89.77	81.75	70.87
	0.00	0.00	100.0	100.0	100.0	100.0	100.0	100.0	100.0
	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
OA (%)	73.34	72.26	**76.27**	75.97	74.16	**79.17**	72.59	70.08	**72.96**
AA (%)	77.94	76.98	**85.64**	79.75	78.15	**82.90**	76.17	74.30	**77.20**
K×100	70.50	69.45	**73.88**	73.43	71.49	**76.93**	69.24	66.39	**69.70**
S (%)	**93.53**	92.37	91.11	**95.70**	93.79	93.03	**91.41**	89.16	88.64
U (%)	0.00	0.00	**58.28**	0.00	0.00	**32.23**	0.00	0.00	**20.00**
H×100	0.00	0.00	**71.09**	0.00	0.00	**47.88**	0.00	0.00	**32.63**

classes as unseen classes, it becomes clear that the mean spectral vector of the seventh class is the most similar to that of the first class in the spectral feature space, and the semantic feature of the seventh class is more similar to that of the

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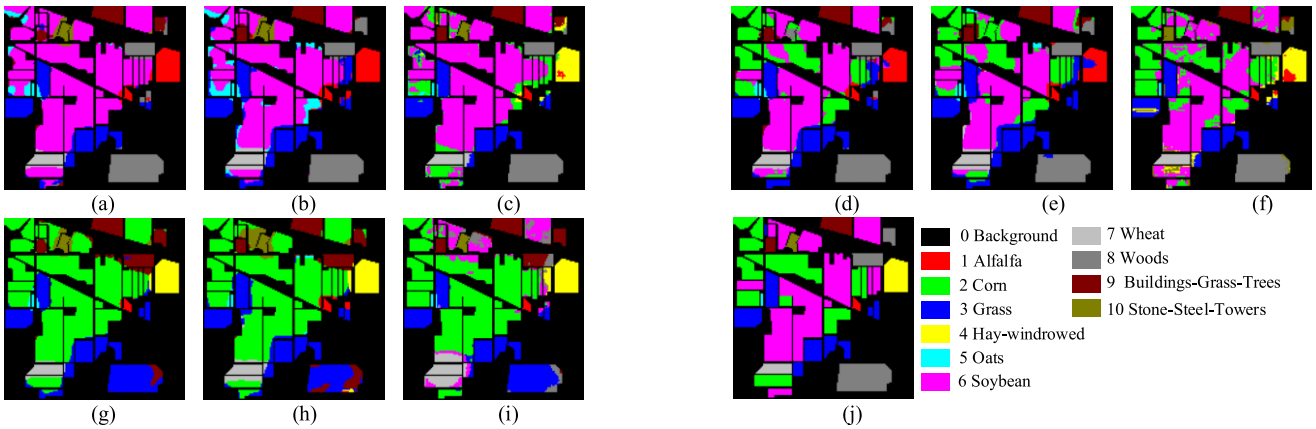


Fig. 21. Classification maps using different methods on the merged Indian Pines dataset. (a)–(c) (second and fourth) CTMixer, DBDA, and SS-VLFMT; (d)–(f) (fourth and tenth) CTMixer, DBDA, and SS-VLFMT; (g)–(i) (sixth and eighth) CTMixer, DBDA, and SS-VLFMT; and (j) ground-truth.

TABLE XVII

CLASSIFICATION RESULTS ON THE MERGED INDIAN PINES DATASET

	(1,9)	(3,10)	(4,10)	(6, 8)	(9,10)	(1,2)
	2.17	100.0	100.0	100.0	100.0	4.35
	68.81	78.59	70.99	81.86	76.00	11.94
	91.65	37.79	87.31	89.93	90.25	84.28
	97.60	100.0	83.89	100.0	100.0	100.0
	100.0	100.0	100.0	100.0	100.0	100.0
	58.18	54.83	75.88	16.87	67.25	91.33
	100.0	100.0	100.0	100.0	100.0	100.0
	99.68	98.39	89.08	13.44	97.27	92.61
	36.27	99.45	75.96	100.0	40.67	99.45
	100.0	11.83	0.00	100.0	2.15	100.0
OA (%)	71.74	68.17	77.89	50.38	76.45	71.53
AA (%)	75.44	78.09	78.31	80.21	77.36	78.40
K×100	65.18	60.63	70.90	38.76	69.77	60.67
S (%)	89.49	91.41	87.40	96.47	91.35	95.96
U (%)	19.22	24.81	41.95	15.15	21.41	8.15
H×100	31.65	39.03	56.69	26.19	34.69	15.02

TABLE XVIII

CLASSIFICATION RESULTS ON THE PAVIA UNIVERSITY DATASET

	(1,4)	(2,9)	(4,5)	(5,8)	(8,9)	(6,8)
	30.18	88.38	91.15	88.59	88.17	95.11
	95.24	21.82	87.97	77.20	90.86	92.51
	81.67	97.31	95.96	99.86	97.74	97.64
	40.11	97.31	23.24	98.00	97.40	84.76
	99.92	100.0	34.13	2.23	100.0	100.0
	97.34	98.46	95.37	97.32	96.43	0.24
	100.0	100.0	100.0	98.32	85.95	100.0
	96.42	97.54	95.49	75.23	40.90	3.83
	100.0	19.22	100.0	100.0	40.76	99.68
OA (%)	81.24	61.50	84.65	82.53	86.61	74.69
AA (%)	82.32	80.00	80.37	81.86	82.02	74.86
K×100	75.32	55.39	80.03	78.01	82.67	65.55
S (%)	95.80	97.00	95.13	94.18	93.79	95.67
U (%)	35.14	20.52	28.68	38.73	40.83	2.03
H×100	51.42	33.88	44.08	54.89	56.89	3.98

first class than to that of the fourth class. A similar situation applies to the fourth class. Consequently, we can infer that the accuracies for these two unseen classes are likely to be higher, a conclusion that is supported by Table XVIII. Similar analyses can be performed for the other examples presented in Table XVIII.

TABLE XIX

CLASSIFICATION RESULTS ON THE MERGED HOUSTON DATASET

	(1,4)	(2,4)	(3,8)	(4,7)	(5,11)	(1,5)
	45.71	84.99	81.57	85.19	84.47	1.36
	100.0	6.31	100.0	99.85	100.0	99.11
	93.63	99.35	23.39	86.19	83.82	99.10
	15.38	75.28	100.0	5.88	96.56	99.18
	93.44	92.13	82.30	92.13	3.69	3.07
	52.05	38.84	74.84	55.58	56.14	62.32
	91.88	85.96	65.75	42.09	85.63	88.47
	83.43	93.62	48.00	88.15	89.81	92.71
	78.02	87.41	95.23	88.81	86.50	88.12
	66.83	65.58	63.02	74.26	62.49	69.26
	100.0	100.0	100.0	100.0	89.95	100.0
	100.0	100.0	100.0	100.0	100.0	99.84
OA (%)	67.44	73.90	74.16	71.17	78.62	68.68
AA (%)	76.70	77.46	77.84	76.51	78.26	75.22
K×100	63.85	71.14	70.98	67.88	76.49	65.61
S (%)	85.93	84.79	86.27	87.02	84.54	89.82
U (%)	30.54	40.80	35.70	23.99	46.82	2.22
H×100	45.07	55.09	50.50	37.60	60.27	4.33

Fig. 20 demonstrates the interclass distances for the merged Houston dataset. When using the second and fourth classes as unseen classes, it is evident that the spectral vector of the fifth class is the most similar to that of the second class in the spectral feature space. However, the semantic feature of the fifth class is more similar to that of the fourth class than to that of the second class. In addition, the nearest neighbor class for the fourth class in the spectral feature space is the 12th class, which corresponds to the fact that the semantic feature of the 12th class is more similar to the fourth class than to the second class. Hence, we can infer that only the accuracy of the fourth class is likely to be higher, and Table XIX confirms this analysis. Similar analyses can be applied to other examples in Table XIX.

In summary, we provide a preliminary analysis of HSI zero-shot classification and guidelines for selecting unseen classes in experiments. However, it is important to acknowledge that this analysis has limitations. For instance, utilizing spatial–spectral features for calculating interclass distances may be more reasonable.

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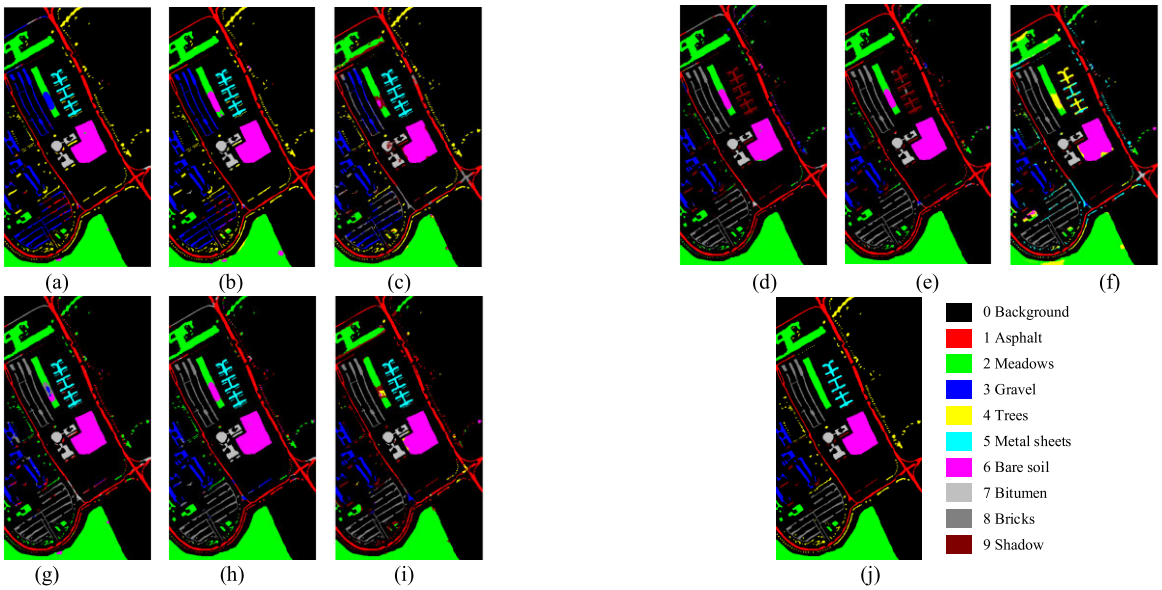


Fig. 22. Classification maps using different methods on the Pavia University dataset. (a)–(c) (eighth and ninth) CTMixer, DBDA, and SS-VLFMT; (d)–(f) (fourth and fifth) CTMixer, DBDA, and SS-VLFMT; (g)–(i) (fourth and ninth) CTMixer, DBDA, and SS-VLFMT; and (j) ground-truth.

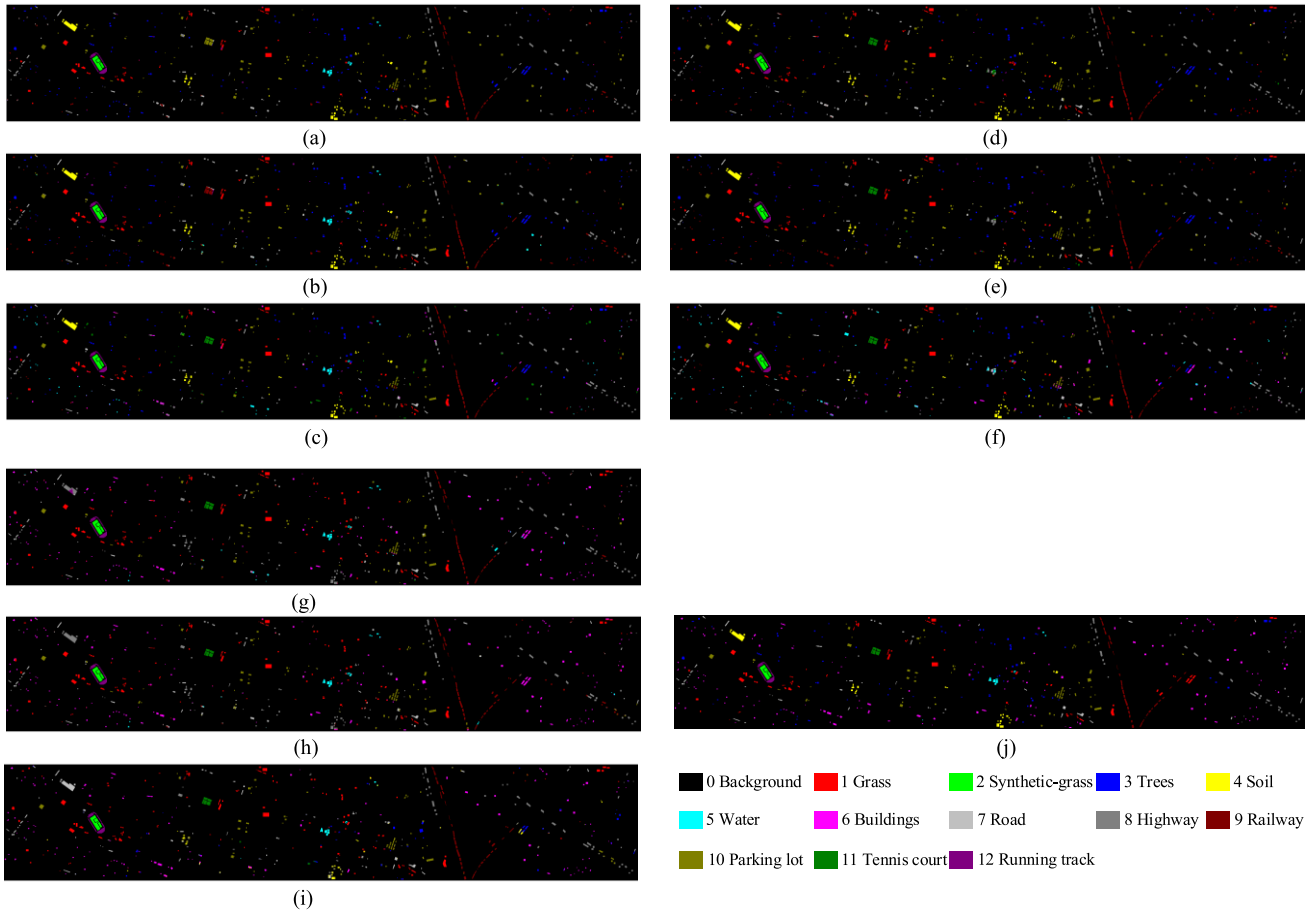


Fig. 23. Classification maps using different methods on the merged Houston dataset. (a)–(c) (sixth and 11th) CTMixer, DBDA, and SS-VLFMT; (d)–(f) (fifth and sixth) CTMixer, DBDA, and SS-VLFMT; (g)–(i) (third and fourth) CTMixer, DBDA, and SS-VLFMT; and (j) ground-truth.

E. Classification Maps of SS-VLFMT

Figs. 21–23 show the classification maps for different methods on the three datasets, according to Tables XIV–XVI. From these maps, one can see that the proposed method can recognize some samples of the unseen classes correctly. For example, when comparing Fig. 21(a)–(c) on the Indian Pines dataset, one can find that the DBDA and CTMixer tend to misclassify the samples of the second class (i.e., corn) as the sixth class (i.e., soybean). We infer that these models are

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overfitting to spatial information because they are close to each other. However, the proposed SS-VLFMT has the ability to classify correctly. When the unseen classes are Bricks and Shadows on the Pavia University dataset [see Fig. 22(a)–(c)], the traditional methods like DBDA and CTMixer misclassify the samples of the class Shadows as a known class from the training data. However, though the samples of class Shadows are close to the other classes' samples, the SS-VLFMT can recognize them correctly.

VII. Conclusion

In this study, we present an attempt to adapt the transformer-based foundation model to HSI classification effectively, leading to the proposed SS-VFMT, SS-VFMT-D, and SS-VLFMT. SS-VFMT can capture spatial and spectral information that modulates the attention scores for better adaption according to the HSI classification task. Besides, SS-VFMT-D further improves the classification performance by patch-level relationship distillation. Based on SS-VFMT, SS-VLFMT further utilizes a language model to make an attempt at GZSL in HSI classification. The proposed methods are then evaluated on three widely used datasets (i.e., Indian Pines, Pavia University, and Houston datasets), and some main conclusions can be drawn from the experimental results as follows:

1) According to the comparative analysis of classification results, the proposed SS-VFMT and SS-VFMT-D can utilize a given transformer-based foundation model to achieve superior performance for the HSI classification task. Meanwhile, the computational burden can be reduced owing to just introducing and tuning a small set of learnable parameters related to the spectral–spatial information of HSI.
2) The ablation experiments show that the proposed SpaEM and SpeEM can both help the foundation transformer adapt to the HSI classification task and obtain better classification results.
3) The intraimage and interimage patch relationship distillation losses are effective for benefiting the training process of SS-VFMT, leading to better classification accuracies.
4) The preliminary results on the GZSL for HSI classification experiments show that the proposed SS-VLFMT demonstrates a certain level of ZSL capability. Zero-shot classification of HSI is a novel and feasible research direction.

This research provides a new direction for further studies to explore the use of foundation models in HSI processing. In the future, the proposed methods can utilize foundation models that are pretrained on remote sensing datasets, leading to further bridging the gap between pretrained knowledge and HSIs.

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